G4NDL4.2 Final State Guide

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Notes:
This documentation was created using geant4.10.00.p02, thus newer versions of geant4 will have a slightly different format, and different class, function, and variable names. However, to my knowledge the format of the final state data files themselves have changed minimally. So only small changes should be required to update this documentation to the current standard.

The starting points in the GEANT4 code of the extraction of the final state data for the four fundamental neutron-isotope interactions

1. G4NeutronHPCapture()
2. G4NeutronHPElastic()
3. G4NeutronHPFission()
4. G4NeutronHPInelastic()

Radiative Capture

There are two directories containing final state files for radiative capture, each with their own method of data extraction

A) FS/:

The data extraction happens in three parts

1. First the InitMean

defines the relationship between outgoing photon energy, the probability of the photon occurring and the fraction of photons that will transfer their kinetic energy to an electron
[repFlag]: refers to the format that the data is presented in
if repFlag can be extracted
[targetMass]: is the mass of the isotope in {/(mass of a neutron)}
Now depending on the repFlag the rest of the data is extracted in the following ways

A) repFlag=1

this format uses a simple interpolation scheme to represent the dependence of photon multiplicity on the incoming neutron kinetic energy, for each possible discrete photon energy
[nDiscrete]: is the number of discrete types of photons that can be emitted for this isotope
For each discrete type of photon the following data is collected

i. [disType]: the photon energy distribution type (continuous [1] or discrete [0]) , determines whether the photon energy will be interpolated or not
ii. [energy]: the energy of the produced photon in {eV}
iii. [total]: the number of incoming neutron kinetic energy points
iv. [nRanges]: defines the number of interpolation ranges (see [1] for explanation of ranges) for the interpolation of incoming neutron kinetic energy and the photon multiplicity.
v. For each range [nRanges], the following data is extracted
    i. [range]: the end point of the range
    ii. [it]: an integer representing the interpolation scheme to be used
vi. For each incoming neutron kinetic energy point [total], the following data is extracted
    i. [x]: the incoming neutron kinetic energy {eV}
    ii. [y]: the photon multiplicity (i.e. the number of photons produced)

B) repFlag=2

This format is a table of photon energy vs the relative probability of the photon being produced at that energy. The effects of different incoming neutron energies on the photon energy are taken care of later in the application of the process to the track
[theInternalConversionFlag]: this number isn't used for anything
[theBaseEnergy]: is the Q-value of the reaction and thus the maximum possible photon energy {eV}
[theInternalConversionFlag]: specifies whether the photon transition fraction data is present in the file
[nGammaEnergies]: specifies the number of photon energy points that there is data for
For each photon energy point [nGammaEnergies], the following data is collected

if theInternalConversionFlag = 1

[theLevelEnergies]: The energy difference between the isotope before the reaction (ground state) and after the photon has been produced (some excited state) {eV}. Photon energy = theBaseEnergy- theLevelEnergies

[theTransitionProbabilities]: The probability of a photon with this energy being produced

if theInternalConversionFlag = 2

[theLevelEnergies]: The energy difference between the isotope before the reaction (ground state) and after the photon has been produced (some excited state) {eV}. Photon energy = theBaseEnergy- theLevelEnergies

[theTransitionProbabilities]: The probability of a photon with this energy being produced

[thePhotonTransitionFraction]: The probability of an electron with the same KE as the photon being released instead of the photon (due to the absorption of the photon by a valence electron)

2. InitAngular

Defines the relationship between incoming neutron kinetic energy and the probability of a photon with a specific angle and energy being produced

[isoFlag]: specifies whether the distribution of photons is isotropic or not

if isoFlag=1

No more data is collected, the momentum direction angles of created photons will be selected from a uniformly random distribution.

If isoFlag!=1

A) if repFlag==2

The code spits out an error since this should not occur

B) [tabulationType]: specifies how the photon angular probability is tabulated, either as Legendre polynomials, or a simple interpolation scheme (in both cases the probability of occurrence is a function of the cosine of the angle)

C) [nDiscrete2]: is the number of discrete photon energies that can be emitted for this isotope, should be equal to nDiscrete

D) [nIso]: is the number of isotropic photons (i.e.: photons where the probability is independent of angle)

E) for each isotropic photon i.e. [nIso], the following data is collected

[theGammas]: The energy of the released photon {eV}

[theShells]: The binding energy of the neutron for a particular shell, this array is not used for anything {eV}

F) for each nonisotropic photon [nDiscrete-nIso], the following data is collected

If tabulationType==1 (i.e. photon angular probability is tabulated as Legendre polynomials which depend on the cosine of the direction angle)

[theGammas]: The energy of the released photon {eV}

[theShells]: The binding energy of the neutron for a particular shell, this array is not used for anything {eV}

[nNeu]: the number of incoming neutron kinetic energy points for this photon

The data in this section is extracted but not used for anything

[nRanges]: defines the number of interpolation ranges (see [1] for explanation of ranges) for the interpolation of incoming neutron kinetic energy and the Legendre polynomials that relate probability of occurrence to the scattering angle

For each range [nRanges], the following data is extracted

[range]: the end point of the range

[it]: an integer representing the interpolation scheme to be used

for each incoming neutron kinetic energy point [nNeu], the following data is collected

[eNeu]: the energy of the incoming neutron {eV}

[nPoly]: the number of the Legendre polynomials used to describe the relationship between cos(theta) (theta is the scattering angle and the probability of occurrence, for this particular case (the domain of a Legendre polynomial is cos(theta) and the range is the probability)

for each Legendre polynomial nPoly, the following data is collected

[coeff]: the coefficient of the Legendre polynomial

If tabulationType=2 (i.e. photon angular probability is tabulated as a simple interpolation scheme)

[theGammas]: The energy of the released photon {eV}

[theShells]: The binding energy of the neutron for a particular shell, this array is not used for anything {eV}

[nNeu]: the number of incoming neutron kinetic energy points for this photon
for each incoming neutron kinetic energy point [nNeu], the following data is collected
i. [eNeu]: the energy of the incoming neutron {eV}
ii. [nProb]: the number of probability points at this energy
iii. [nRanges]: defines the number of interpolation ranges (see [1] for explanation of ranges) for the interpolation of cosTheta and probability of occurrence
iv. For each range [nRanges], the following data is extracted
a) [range]: the end point of the range
b) [it]: an integer representing the interpolation scheme to be used
v. for each probability point [nProb], the following data is collected
a) [cosTheta]: the cosine of the direction angle of the photon
b) [probDist]: the probability of the photon being released under the previous circumstances

3. InitEnergies

In this section the data that defines the continuous photon energy distributions is collected
if disType[i]!=1 for any of the photon energy points extracted in InitMean
no more data is collected, the energy of created photons will not be interpolated and will simply be the closest discrete photon energy point
else
[nPartials]: the number of distinct states that the nuclei will assume after absorbing the neutron
for each distinct states that the nuclei will assume [nPartial], the following data is extracted
a) [dummy]: useless number
b) [total]: number of points for interpolating the probability of this state occurring
c) [nRanges]: defines the number of interpolation ranges (see [1] for explanation of ranges) for the interpolation of incoming neutron kinetic energy and the probability of the nucleus entering this distinct state.
d) For each range [nRanges], the following data is extracted
[range]: the end point of the range
[it]: an integer representing the interpolation scheme to be used
e) for each interpolation point [total], the following data is extracted
[x]: the incoming neutron kinetic energy {eV}
[y]: the probability of occurrence of this state at the incoming neutron kinetic energy
f) [nen]: number of incoming neutron kinetic energy points for the interpolation of the photon probability distribution
g) [nRanges]: defines the number of interpolation ranges for the interpolation of incoming neutron kinetic energy and the probability of the photon being created with a specific energy.
h) For each range [nRanges], the following data is extracted
[range]: the end point of the range
[it]: an integer representing the interpolation scheme to be used
i) For each neutron energy point [nen], the following data is extracted
[e]: the incoming neutron kinetic energy {eV}
[neg]: the number of photon energy points
[nRanges]: defines the number of interpolation ranges for the interpolation of the photon energy and the probability of the photon occurring.
For each range [nRanges], the following data is extracted
[range]: the end point of the range
[it]: an integer representing the interpolation scheme to be used
For each photon energy point [neg], the following data is extracted
[eg]: the photon energy {eV}
[pg]: the probability of occurrence

B) FSMF6/:

If this data is present it is used instead of the previously defined data for the capture reaction
this data defines the angular-energy distribution for photons
[targetMass]: the mass of the isotope {/(mass of a neutron)}
[frameFlag]: the relativistic frame that the data is taken with respect to
[nProducts]: the number of states that the nucleus can go into after the absorption of the neutron
for each states of the nucleus [nProducts], the following data is extracted

1. [theMassCode]: the 5 digit ZZAAA code for the out going particle code

2. [theMass]: not used for anything
3. [theIsomerFlag]: not used for anything
4. [theDistLaw]: specifies the distribution of the photons, i.e. isotropic, continuous angular-energy dist, discrete 2 body scattering
5. [theGroundStateQValue]: not used for anything, the energy difference between is excited state and the ground state, i.e. the theoretical Q-value of the reaction {eV}
6. [theActualStateQValue]: the energy difference between the excited state after absorption and the ground state before absorption, i.e. the actual Q-value of the reaction, should be very close to the theoretical {eV}
7. [total]: the total number of incoming neutron kinetic energy and out-going photon multiplicity (the number of photons released) points.
8. [nRanges]: defines the number of interpolation ranges for the interpolation between the incoming neutron kinetic energy and out-going photon multiplicity.
9. For each range [nRanges], the following data is extracted
   A) [range]: the end point of the range
   B) [it]: an integer representing the interpolation scheme to be used
10. for each out-going photon multiplicity point [total], the following data is extracted
    A) [x]: the incoming neutron kinetic energy {eV}
    B) [y]: the out-going photon multiplicity
11. if theDistLaw=0,3
    A) An isotropic photon angular distribution is used
    B) No data is collected from the file, instead uniform random distributions are used to sample the angles
12. if theDistLaw=1
    A) A continuous energy and angular distribution is used
    B) [theTargetCode]: is the ZZAAA integer that specifies the nucleus
    C) [theAngularRep]: is an integer that specifies how the angular distribution is to be sampled
    D) [theInterpolation]: specifies the interpolation scheme to be used for the interpolation between probability of the photon being created with a certain energy and angle, and the incoming neutron kinetic energy
    E) [nEnergy]: the number of incoming neutron kinetic energy points
    F) for each incoming neutron kinetic energy point [nEnergy], the following data is extracted
       a) [theEnergy]: the incoming neutron kinetic energy {eV}
       b) [nEnergies]: the number of photon energy points
       c) [nDiscreteEnergies]: the number of photon energy points that have discrete energies i.e. the number of points that are not part of the continuous photon energy dist (the photon can only occur at the energy point, there is no interpolation in-between the points), thus nDiscreteEnergies<= nEnergies, and nEnergies- nDiscreteEnergies is the number of photon energy points that are part of the continuous energy dist.
       d) [nAngularParameters]: the number angle probability points for each photon energy point
       e) for each photon energy point [nEnergies], the following data is extracted
          i. [sEnergy]: the out-going photon energy {eV}
          ii. for each angle probability point nAngularParameters, the following data is extracted
              i. [y]: the probability of the occurrence of the photon (the photon meaning at the specified isotope state, incoming neutron kinetic energy, photon energy, and angle). Note the angle corresponding to the probability is based off the probability array index, i.e. each array index in the probability array is a set angle apart.
13. if theDistLaw=2,4
    A) a discrete 2 body recoil distribution is used, this type of distribution uses a set of Legendre polynomials to describe the angular dependence of the probability of a photon occurring for each photon energy, there are multiple photon energies for each incoming neutron kinetic energy
    B) [nEnergy]: the number of neutron energy points
    C) [nRanges]: defines the number of interpolation ranges for the interpolation of between the Legendre polynomial sets.
    D) For each range [nRanges], the following data is extracted
       a) [range]: the end point of the range
       b) [it]: an integer representing the interpolation scheme to be used
    E) for each neutron energy point [nEnergy], the following data is collected
       a) [energy]: the incoming neutron kinetic energy {eV}
       b) [aRep]: the interpolation scheme between the Legendre coefficients

c) [nCoeff]: the number of Legendre polynomial coefficients
d) if aRep>0
    i. then the total number of Legendre coefficients is 2xnCoeff
    ii. for the total number of Legendre coefficients, the following data is extracted
        i. [y]: is Legendre polynomial coefficient
e) if aRep=0 then
    i. then the total number of Legendre coefficients is nCoeff
    ii. for the total number of Legendre coefficients, the following data is extracted
        i. [y]: is Legendre polynomial coefficient

14. if theDistLaw=5
    A) does nothing, has not been coded yet
15. if theDistLaw=6
    A) an n-body phase space distribution is used
    B) [theTotalMass]: the mass of the isotope divided by the mass of a neutron {MeV}
    C) [theTotalCount]: the number of particles involved in the reaction
16. if theDistLaw=7
    A) an experimental parameterized energy and angular model
    B) [nEnergies]: the number of incoming neutron energies
    C) [nRanges]: defines the number of interpolation ranges for the interpolation of between incoming neutron kinetic energy and the probability of the photon being shot in at a specific angle.
    D) For each range [nRanges], the following data is extracted
        a) [range]: the end point of the range
        b) [it]: an integer representing the interpolation scheme to be used
    E) for each incoming neutron kinetic energy [nEnergies], the following data is collected
        a) [theEnergies]: the incoming neutron kinetic energy {eV}
        b) [nCosTh]: the number of angle points for this incoming neutron kinetic energy
        c) [nRanges]: defines the number of interpolation ranges for the interpolation of between the photon angular direction and the probability of the photon having a specific energy.
        d) For each range [nRanges], the following data is extracted
            i. [range]: the end point of the range
            ii. [it]: an integer representing the interpolation scheme to be used
        e) for each angle point [nCosTh], the following data is extracted
            i. [label]: this is the cosine of the angle
            ii. [total]: the number of photon energy points
            iii. [nRanges]: defines the number of interpolation ranges for the interpolation of between the photon energy and the probability of the photon occurring.
            iv. For each range nRanges, the following data is extracted
                i. [range]: the end point of the range
                ii. [it]: an integer representing the interpolation scheme to be used
            v. for each photon energy point [total], the following information is extracted
                i. [x]: the photon energy {eV}
                ii. [y]: the probability of occurrence

Elastic Scattering

1. The first number [repFlag] refers to the format that the data is presented in
2. [targetMass]: is the mass of the isotope in {/(mass of a neutron)}
3. [frameFlag]: the reference frame that the data was measured with respect to (1=LAB, 2=CMS)
4. Now depending on the repFlag the rest of the data is extracted in the following ways
5. repFlag=0
    a) When this format is called it means there is no data to extract
    b) The frameFlag is collected again and then other operations are done
    c) This format is only meant for isotopes that are missing data for their elastic scattering or the probability of elastic scattering occurring for the isotope is not significant
6. repFlag=1
    a) This format uses a Legendre polynomial to describe the relationship between scattering angle and probability of

occurrence for each incoming neutron kinetic energy.
   b) [nEnergy]: refers to the number of incoming neutron kinetic energy points
   c) [nRanges]: defines the number of interpolation ranges for interpolating between incoming neutron kinetic energy and Legendre polynomials.
   d) For each interpolation range the following two numbers are collected
      i. [range]: The end point of the range
      ii. [it]: The interpolation scheme for the range
   e) For each energy point [nEnergy], the following data is collected
      i. [temp]: the temperature the isotope was at when the data was collected, not used for anything
      ii. [energy]: the relative kinetic energy of the incoming particle with respect to the target isotope
      iii. [tempdep]: a temperature dependence flag that is no longer used by the code
      iv. [nLegendre]: the number of Legendre polynomials defining the dependence of the scattering probability on the angle on probability for this incoming neutron kinetic energy
      v. For each Legendre polynomial [nLegendre] the following data is collected
         1. [coeff]: the coefficient for the current Legendre polynomial
7. repFlag=2
   a) this format uses a set of interpolation schemes, one for each incoming neutron angle, to interpolate between scattering angle and probability
   b) [nEnergy]: refers to the number of incoming neutron kinetic energy point
   c) [nRanges]: defines the number of interpolation ranges, for interpolating between incoming neutron kinetic energy and the angular probability distribution.
   d) For each interpolation range the following two numbers are collected
      i. [range]: the end point of the range
      ii. [it]: The interpolation scheme for the range
   e) For each incoming neutron kinetic energy point [nEnergy] the following data is collected
      i. [temp]: the temperature the isotope was at when the data was collected, currently not used for anything
      ii. [energy]: the relative kinetic energy of the incoming particle with respect to the target isotope
      iii. [tempdep]: a temperature dependence flag that is no longer used by the code
      iv. [nPoints]: the number of probability points for this energy point
      v. [nRanges]: defines the number of interpolation ranges, for interpolating between the cosine of the scattering and the probability of occurrence.
      vi. For each interpolation range the following two numbers are collected
         1. [range]: the end point of the range
         2. [it]: The interpolation scheme for the range
      vii. For each probability point [nPoints], the following data is collected
         1. [costh]: the cosine of the scattering angle (the angle between the original trajectory of the neutron and the final trajectory of the neutron)
         2. [prob]: the probability of elastic scattering occurring at this energy and angle
8. repFlag=3
   a) this format is simply a mixture of the previous two
   b) [nEnergy_Legendre]: the number of incoming neutron kinetic energy points for the Legendre polynomial
   c) The fifth number [nRanges] defines the number of interpolation ranges.
   d) For each interpolation range [nRanges], the following two numbers are collected
      i. [range]: the end point of the range
      ii. [it]: The interpolation scheme for the range
   e) For each Legendre energy point [nEnergy_Legendre], the following data is collected
      i. [temp]: the temperature the isotope was at when the data was collected
      ii. [energy]: the relative kinetic energy of the incoming particle with respect to the target isotope
      iii. [tempdep]: a temperature dependence flag that is no longer used by the code
      iv. [nLegendre]: the number of Legendre polynomial defining the dependence of the scattering probability on scattering angle for this energy point
      v. For each Legendre polynomial [nLegendre], the following data is collected
         1. [coeff]: the coefficient for the current Legendre polynomial
   f) [nEnergy_Prob]: is the number of incoming neutron kinetic energy points for the tabular interpolation section
   g) [nRanges]: defines the number of interpolation ranges, for interpolating between the incoming neutron kinetic energy and the angular probability distribution.
   h) For each interpolation range the following two numbers are collected

i. [range]: the end point of the range
ii. [it]: The interpolation scheme for the range

i) For each energy probability point the following data is collected
i. [temp]: the temperature the isotope was at when the data was collected
ii. [energy]: the relative kinetic energy of the incoming particle with respect to the target isotope
iii. [tempdep]: a temperature dependence flag that is no longer used by the code
iv. [nPoints]: the number of probability points for this energy point
v. [nRanges]: defines the number of interpolation ranges, for interpolating between the scattering angle and the probability of occurrence.
vi. For each interpolation range the following two numbers are collected
1. [range]: the end point of the range
2. [it]: The interpolation scheme for the range
vii. For each probability point the following data is collected
1. [costh]: the cosine of the scattering angle (the angle between the original trajectory of the neutron and the final trajectory of the neutron)
2. [prob]: the probability of elastic scattering occurring at this energy and angle

Fission

The data for each isotope is split up into 7 files each contained in a separate directory
Each directory contains a certain type of data and has the following format

1. FS/

A) The data collected in this folder represents the total/average fission reaction, MT=18, and it is used to determine the created photons, the number of prompt and delayed neutrons created, the created delayed neutrons, and the created prompt neutrons if the other fission directories do not contain prompt neutron data for this isotope.
B) while there is data left in the file, the following data is extracted
a) [infoType]: specifies how the data is to be extracted
b) [dataType]: specifies what type of data is stored in this section of the file and thus the extraction method
c) if infoType=1
if dataType=4
theNeutronAngularDis.Init()
i. The data extracted in this section defines the probability that a hadron with a particular angle will be produced by an incoming neutron with a particular energy if fission occurs
ii. [theAngularDistributionType]: determines how the angular distribution of the produced hadrons are to be represented and subsequently the format of the data to be extracted
iii. [targetMass]: the mass of the isotope in {/(mass of a neutron)}
iv. [frameFlag]: the reference frame that the data was measured with respect to (1=LAB, 2=CMS)
v. if theAngularDistributionType = 0
1. the angular distribution of the produced hadrons is to be represented by an isotropic distribution which means that no further data is necessary
vi. if theAngularDistributionType = 1
1. The angular distribution of the produced hadrons (remember the incoming is absorbed) is to be represented by Legendre polynomial sets which represent the probability angular relationship. There is a set of Legendre polynomials for each incoming neutron kinetic energy and each of the polynomials in a set are centered around a particular exit angle (meaning that the closer the angle is to the angle the Legendre polynomial is centered on, the better the approximation of that polynomial becomes). The coefficients of the Legendre polynomials are dependent on energy of the incoming neutrons and the angle of the exiting hadron.
2. [nEnergy]: the number of incoming neutron kinetic energy points and also the number of Legendre polynomial sets
3. [nRanges]: defines the number of interpolation ranges for the interpolation of between incoming neutron kinetic energy and the Legendre polynomial sets.
4. For each range nRanges, the following data is extracted
a) [range]: the end point of the range
b) [it]: an integer representing the interpolation scheme to be used
5. for each Legendre polynomial set nEnergy, the following data is extracted

a) [temp]: the temperature of the isotope
b) [energy]: the energy of the incoming neutron {eV}
c) [tempdep]: not used for anything
d) [nLegendre]: the number of Legendre polynomials in the set and also the number of exit angle probability points
e) for each Legendre polynomial nLegendre, the following data is extracted
i. [coeff]: the coefficient of the Legendre polynomial
ii. note: the angle that the polynomial is centered around is based off the indices of the polynomial in the set

vii. if theAngularDistributionType = 2
1. The angular distribution is represented by a 3D table of incoming neutron kinetic energy, exit angle and probability
2. [nEnergy]: the number of incoming neutron kinetic energy points
3. [nRanges]: defines the number of interpolation ranges for the interpolation of between the incoming neutron kinetic energy points and the exiting neutrons angular probability distribution.
4. For each range nRanges, the following data is extracted
a) [range]: the end point of the range
b) [it]: an integer representing the interpolation scheme to be used
5. for each incoming neutron kinetic energy point nEnergy, the following data is extracted
a) [temp]: the temperature of the isotope that the data was taken at
b) [energy]: the energy of the incoming neutron kinetic energy {eV}
c) [tempdep]: not used for anything
d) [neg]: the number of exit angle points for this incoming neutron kinetic energy
e) [nRanges]: defines the number of interpolation ranges for the interpolation of between the exit angle points.
f) For each range nRanges, the following data is extracted
i. [range]: the end point of the range
ii. [it]: an integer representing the interpolation scheme to be used
g) for each exit angle point neg, the following data is extracted
i. [eg]: the exit angle of the hadron
ii. [pg]: the probability of this hadron being created

2. if dataType=5
A) thePromptNeutronEnDis.Init
i. The data collected in this section defines the energy distribution of the produced prompt neutrons using various models. Each model contains an incoming neutron kinetic energy and model weight (the model accuracy at a particular incoming neutron kinetic energy) interpolation scheme which allows the process to stochastically select a model to be used for a particular incoming neutron
ii. [dummy]: does nothing
iii. [theNumberOfPartials]: the number of partials that will make up interpolation scheme between incoming neutron kinetic energy and outgoing hadron energy
iv. for each partial theNumberOfPartials, the following data will be extracted
1. [theRepresentationType]: Specifies how the data has been formatted
2. if theRepresentationType=1
a) [total]: the number of incoming neutron kinetic energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
b) [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.
c) For each range nRanges, the following data is extracted
i. [range]: the end point of the range
ii. [it]: an integer representing the interpolation scheme to be used
d) for each incoming neutron kinetic energy and representation weight point total, the following data is extracted
i. [x]: the incoming neutron kinetic energy {eV}
ii. [y]: the representation weight, aka the probability of this representation being used
e) [nDistFunc]: the number of interpolation schemes used to describe the relationship between

out-going hadron energy and the probability of the hadron leaving with that energy. there is one interpolation scheme per incoming neutron kinetic energy so this is also the number of incoming neutron kinetic energy points

f) [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and the probability of a hadron being ejected with a particular energy.
g) For each range nRanges, the following data is extracted
   i. [range]: the end point of the range
   ii. [it]: an integer representing the interpolation scheme to be used
h) For each incoming neutron kinetic energy point nDistFunc, the following data is extracted
   i. [currentEnergy]: the incoming neutron kinetic energy {eV}
   ii. [total]: the number of outgoing hadron energy and probability of occurrence points
   iii. [nRanges]: defines the number of interpolation ranges for the interpolation between outgoing hadron energy and the probability of occurrence.
   iv. For each range nRanges, the following data is extracted
   [range]: the end point of the range
   [it]: an integer representing the interpolation scheme to be used
   v. For each outgoing hadron energy point total, the following data is extracted
   [x]: the out-going hadron energy {eV}
   [y]: the probability of occurrence

3. if theRepresentationType=5
   a) For neatness the following is organised into three parts
   b) theFractionalProb
      i. [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
      ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.
      iii. For each range nRanges, the following data is extracted
      [range]: the end point of the range
      [it]: an integer representing the interpolation scheme to be used
      iv. for each energy and representation weight point total, the following data is extracted
      [x]: the energy of the incoming neutron {eV}
      [y]: the representation weight, aka the probability of this representation being used
   c) theThetaDist
      i. [total]: the total number of incoming neutron kinetic energy and average outgoing hadron energy points
      ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and average out-going neutron energy.
      iii. For each range nRanges, the following data is extracted
      [range]: the end point of the range
      [it]: an integer representing the interpolation scheme to be used
      iv. for each incoming neutron kinetic energy and maximum out-going hadron energy point total, the following data is extracted
      [x]: the incoming neutron kinetic energy {eV}
      [y]: the average out-going hadron energy {eV}
   d) theXDist
      i. [total]: the number of multiplier and weight points. The weights are the probability of the multiplier being randomly selected. When an outgoing hadron energy is called for, the average outgoing hadron energy is interpolated from the ThetaDist data and a randomly selected multiplier is applied to it.
      ii. [nRanges]: defines the number of interpolation ranges for the interpolation between multiplier and weight points.
      iii. For each range nRanges, the following data is extracted
      [range]: the end point of the range
      [it]: an integer representing the interpolation scheme to be used
      iv. for each energy and representation weight point total, the following data is extracted

[x]: the multiplier
[y]: the weight

4. if theRepresentationType=7,9
   a) theFractionalProb
      i. [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
      ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.
      iii. For each range nRanges, the following data is extracted
      [range]: the end point of the range
      [it]: an integer representing the interpolation scheme to be used
      iv. for each energy and representation weight point total, the following data is extracted
      [x]: the energy of the incoming neutron {eV}
      [y]: the representation weight, aka the probability of this representation being used
   b) theThetaDist
      i. [total]: the total number of incoming neutron kinetic energy and average outgoing hadron energy points
      ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and average out-going neutron energy.
      iii. For each range nRanges, the following data is extracted
      [range]: the end point of the range
      [it]: an integer representing the interpolation scheme to be used
      iv. for each incoming neutron kinetic energy and maximum out-going hadron energy point total, the following data is extracted
      [x]: the incoming neutron kinetic energy
      [y]: the average out-going hadron energy
5. if theRepresentationType=11
   a) theFractionalProb
      i. [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
      ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.
      iii. For each range nRanges, the following data is extracted
      [range]: the end point of the range
      [it]: an integer representing the interpolation scheme to be used
      iv. for each energy and representation weight point total, the following data is extracted
      [x]: the energy of the incoming neutron {eV}
      [y]: the representation weight, aka the probability of this representation being used
   b) theApar
      i. [total]: the total number of incoming neutron kinetic energy and parameter A of the Watt Spectrum points
      ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and parameter A of the Watt Spectrum.
      iii. For each range nRanges, the following data is extracted
      [range]: the end point of the range
      [it]: an integer representing the interpolation scheme to be used
      iv. for each incoming neutron kinetic energy and parameter A of the Watt Spectrum point total, the following data is extracted
      [x]: the incoming neutron kinetic energy {eV}
      [y]: the parameter A of the Watt Spectrum {eV}
   c) theBpar
      i. [total]: the total number of incoming neutron kinetic energy and parameter B of the Watt Spectrum points

ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and parameter B of the Watt Spectrum.
iii. For each range nRanges, the following data is extracted
[range]: the end point of the range
[it]: an integer representing the interpolation scheme to be used
iv. for each incoming neutron kinetic energy and parameter A of the Watt Spectrum point total, the following data is extracted
[x]: the incoming neutron kinetic energy {eV}
[y]: the parameter B of the Watt Spectrum {MeV}

6. if theRepresentationType=12
   a) theFractionalProb
      i. [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
      ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.
      iii. For each range nRanges, the following data is extracted
      [range]: the end point of the range
      [it]: an integer representing the interpolation scheme to be used
      iv. for each energy and representation weight point total, the following data is extracted
      [x]: the energy of the incoming neutron {eV}
      [y]: the representation weight, aka the probability of this representation being used
   b) [theAvarageKineticPerNucleonForLightFragments]: the average kinetic energy of a light fragment per nucleon in the fragment {eV}
   c) [theAvarageKineticPerNucleonForHeavyFragments]: the average kinetic energy of a heavy fragment per nucleon in the fragment {eV}
   d) theMaxTemp
      i. [total]: the total number of incoming neutron kinetic energy and the max fission fragment temperature points
      ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and max fission fragment temperature.
      iii. For each range nRanges, the following data is extracted
      [range]: the end point of the range
      [it]: an integer representing the interpolation scheme to be used
      iv. for each incoming neutron kinetic energy and parameter A of the Watt Spectrum point total, the following data is extracted
      [x]: the incoming neutron kinetic energy {eV}
      [y]: the max fission fragment temperature

3. if dataType=12
   A) theFinalStatePhotons.InitMean
   B) The data collected in this section defines the distribution of photon multiplicity and incoming neutron kinetic energy
      i. [repFlag]: refers to the format that the data is presented in
      ii. if repFlag can be extracted
         1. [targetMass]: is the mass of the isotope in {/(mass of a neutron)}
         2. Now depending on the repFlag the rest of the data is extracted in the following ways
         3. repFlag=1
            a) this format uses a simple interpolation scheme to represent the dependence of photon multiplicity on the incoming neutron kinetic energy, for each possible discrete photon emission
            b) [nDiscrete]: is the number of discrete types of photons that can be emitted for this isotope
            c) For each discrete type of photon the following data is collected
               i. [disType]: the photon energy distribution type (i.e. continuous, ...) , is used to determine how the process is to use the data when it is interpolating from it
               ii. [energy]: the energy of the produced photon {eV}
               iii. [total]: the number of incoming neutron kinetic energy points

iv. [nRanges]: defines the number of interpolation ranges (see [1] for explanation of ranges) for the interpolation of incoming neutron kinetic energy and the photon multiplicity.
v. For each range [nRanges], the following data is extracted
[range]: the end point of the range
[it]: an integer representing the interpolation scheme to be used
vi. For each incoming neutron kinetic energy point the following data is extracted
[x]: the incoming neutron kinetic energy {eV}
[y]: the photon multiplicity

4. repFlag=2
   a) This format is a table of photon energy vs the relative probability of the photon being produced at that energy. The effects of different incoming neutron energies on the photon energy are taken care of later in the application of the process to the track
   b) [theInternalConversionFlag]: isn't used for anything
   c) [theBaseEnergy]: is the base photon energy used in the creation of photons by the fission process
   d) [theInternalConversionFlag]: specifies whether the photon transition fraction data is present in the file
   e) [nGammaEnergies]: specifies the number of photon energy points that there is data for
   f) For each photon energy point the following data is collected
      i. if theInternalConversionFlag = 1
      [theLevelEnergies]: The energy difference between the isotope before the reaction (ground state) and after the photon has been produced (some excited state) {eV}. Photon energy = theBaseEnergy- theLevelEnergies
      [theTransitionProbabilities]: The probability of a photon with this energy being produced
      ii. if theInternalConversionFlag = 2
      [theLevelEnergies]: The energy difference between the isotope before the reaction (ground state) and after the photon has been produced (some excited state) {eV}. Photon energy = theBaseEnergy- theLevelEnergies
      [theTransitionProbabilities]: The probability of a photon with this energy being produced
      [thePhotonTransitionFraction]: The probability of an electron with the same KE as the photon being released instead of the photon

4. if dataType=14
   A) theFinalStatePhotons.InitAngular
   B) The data collected in this section defines the angular distribution of the created photons (i.e. the dependence of a photon being created at a particular angle on incoming neutron kinetic energy and photon energy
      5. [isoFlag]: specifies whether the distribution of photons is isotropic or not
      6. if isoFlag=1
         A) No more data is collected, the momentum direction angles of created photons will be selected from a uniformly random distribution.
      7. If isoFlag!=1
         A) [tabulationType]: specifies how the photon angular probability is tabulated, either as Legendre polynomials, or a simple interpolation scheme (in both cases the probability of occurrence is a function of the cosine of the angle)
         B) [nDiscrete2]: is the number of discrete types of photons that can be emitted for this isotope, should be equal to nDiscrete
         C) [nIso]: is the number of isotropic photons (i.e.: equal probability photons)
         D) for each isotropic photon i.e. nIso, the following data is collected
            i. [theGammas]: The energy of the released photon {eV}
            ii. [theShells]: this array is not used for anything
         E) for each nonisotropic photons (nDiscrete-nIso), the following data is collected
            i. If tabulationType==1 (i.e. photon angular probability is tabulated as Legendre polynomials which depend on the cosine of the direction angle)
               i. [theGammas]: The energy of the released photon {eV}
               2. [theShells]: this array is not used for anything
               3. [nNeu]: the number of incoming neutron kinetic energy points for this photon

4. The data in this section is extracted but not used for anything
    a) [nRanges]: defines the number of interpolation ranges (see [1] for explanation of ranges) for the interpolation of incoming neutron kinetic energy and the Legendre polynomials that relate probability of occurrence to the scattering angle
    b) For each range [nRanges], the following data is extracted
        i. [range]: the end point of the range
        ii. [it]: an integer representing the interpolation scheme to be used
    5. for each incoming neutron kinetic energy point nNeu, the following data is collected
        a) [eNeu]: the energy of the incoming neutron {eV}
        b) [nPoly]: the number of the Legendre polynomials used to describe the relationship between cos(theta) (theta is the scattering angle and the probability of occurrence, for this particular case (the domain of a Legendre polynomial is cos(theta) and the range is the probability)
        c) for each Legendre polynomial nPoly, the following data is collected
            i. [coeff]: the coefficient of the Legendre polynomial

ii. If tabulationType=2 (i.e. photon angular probability is tabulated as a simple interpolation scheme)
    1. [theGammas]: The energy of the released photon {eV}
    2. [theShells]: this array is not used for anything
    3. [nNeu]: the number of incoming neutron kinetic energy points for this photon
    4. for each incoming neutron kinetic energy point nNeu, the following data is collected
        a) [eNeu]: the energy of the incoming neutron {eV}
        b) [nProb]: the number of probability points at this energy
        c) [nRanges]: defines the number of interpolation ranges (see [1] for explanation of ranges) for the interpolation of cosTheta and probability of occurrence
        d) For each range [nRanges], the following data is extracted
            i. [range]: the end point of the range
            ii. [it]: an integer representing the interpolation scheme to be used
        e) for each probability point nProb, the following data is collected
            i. [cosTheta]: the cosine of the direction angle of the photon
            ii. [probDist]: the probability of the photon being released under the previous circumstances

8. if dataType=15
    A) theFinalStatePhotons.InitEnergies
    B) The data collected in this section describes the relationship between incoming neutron kinetic energy and outgoing photon energy, when the photon energy is not discrete.
    C) if disType[i]!=1 for any of the photon energy points extracted in InitMean
        i. no more data is collected, the energy of created photons will not be interpolated and will simply be the closest discrete photon energy point
    D) else
        i. [nPartials]: the number of distinct states that the nuclei will assume after absorbing the neutron
        ii. for each distinct states that the nuclei will assume nPartial, the following data is extracted
            1. [dummy]: useless number
            2. [total]: number of points for interpolating the probability of this state occurring
            3. [nRanges]: defines the number of interpolation ranges (see [1] for explanation of ranges) for the

interpolation of incoming neutron kinetic energy and the probability of the nucleus entering this distinct state.
4. For each range nRanges, the following data is extracted
   a) [range]: the end point of the range
   b) [it]: an integer representing the interpolation scheme to be used
5. for each interpolation point total, the following data is extracted
   a) [x]: the incoming neutron kinetic energy {eV}
   b) [y]: the probability of occurrence of this state at the incoming neutron kinetic energy
6. [nen]: number of neutron energy points for the interpolation of the photon probability distribution
7. [nRanges]: defines the number of interpolation ranges for the interpolation of incoming neutron kinetic energy and the probability of the photon energy.
8. For each range nRanges, the following data is extracted
   a) [range]: the end point of the range
   b) [it]: an integer representing the interpolation scheme to be used
9. For each neutron energy point nen, the following data is extracted
   a) [e]: the incoming neutron kinetic energy {eV}
   b) [neg]: the number of photon energy points
   c) [nRanges]: defines the number of interpolation ranges for the interpolation of the photon energy and the probability of the photon occurring.
   d) For each range nRanges, the following data is extracted
      i. [range]: the end point of the range
      ii. [it]: an integer representing the interpolation scheme to be used
   e) For each photon energy point neg, the following data is extracted
      i. [eg]: the photon energy {eV}
      ii. [pg]: the probability of occurrence

d) if infoType=2
   1. if dataType=1
      A) theFinalStateNeutrons.InitMean
      B) The data collected in this section relates the total neutron production to the incoming neutron kinetic energy
         i. [targetMass]: the mass of the isotope {/(mass of a neutron)}
         ii. [iflag]: specifies whether the total neutron multiplicity will be interpolated using a simple scheme or using a polynomial expansion
         iii. if iflag!=1
            1. [total]: the total number of incoming neutron kinetic energy and outgoing total neutron multiplicity points
            2. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and outgoing total neutron multiplicity.
            3. For each range nRanges, the following data is extracted
               a) [range]: the end point of the range
               b) [it]: an integer representing the interpolation scheme to be used
            4. for each incoming neutron kinetic energy and outgoing total neutron multiplicity point total, the following data is extracted
               a) [x]: the incoming neutron kinetic energy {eV}
               b) [y]: the outgoing total neutron multiplicity
         iv. if iflag=1
            1. [nPoly]: the number of terms in the polynomial expansion
            2. for each polynoimal term nPoly, the following data is extracted
               a) [theCoeff]: the coefficient of the term (order in which these are extracted determines what the order of the term they belong to is)

e) if infoType=3
   1. if dataType=1
      A) theFinalStateNeutrons.InitDelayed
      B) this section defines the relationship between incoming neutron kinetic energy and outgoing delayed neutron multiplicity
         i. [targetMass]: the mass of the isotope {/(mass of a neutron)}

ii. [iflag]: specifies whether the delayed neutron multiplicity will be constant or interpolated from the incoming neutron kinetic energy
iii. [total]: the number of delayed neutron time constants
iv. for each delayed neutron time constant [total], the following data is extracted
    1. [y]: the delay time constant {1/s}
v. if iflag!=2
    1. [theSpontDelayed]: the multiplicity (i.e. number) of delayed neutrons that shall be created for all incoming neutron energies, for this isotope.
vi. if iflag=2
    1. [total]: the total number of incoming neutron kinetic energy and outgoing delayed neutron multiplicity points
    2. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and outgoing delayed neutron multiplicity.
    3. For each range nRanges, the following data is extracted
        a) [range]: the end point of the range
        b) [it]: an integer representing the interpolation scheme to be used
    4. for each incoming neutron kinetic energy and outgoing delayed neutron multiplicity point total, the following data is extracted
        a) [x]: the incoming neutron kinetic energy {eV}
        b) [y]: the outgoing delayed neutron multiplicity

2. if dataType=5
    A) theDelayedNeutronEnDis.Init
    B) The data collected in this section defines the energy distribution of the produced delayed neutrons using various models. Each model contains an incoming neutron kinetic energy and model weight (the model accuracy at a particular incoming neutron kinetic energy) interpolation scheme which allows the process to stochastically select a model to be used for a particular incoming neutron
        i. [dummy]: does nothing
        ii. [theNumberOfPartials]: the number of partials that will make up interpolation scheme between incoming neutron kinetic energy and outgoing hadron energy
        iii. for each partial theNumberOfPartials, the following data will be extracted
            1. [theRepresentationType]: Specifies how the data has been formatted
            2. if theRepresentationType=1
                a) [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
                b) [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.
                c) For each range nRanges, the following data is extracted
                    i. [range]: the end point of the range
                    ii. [it]: an integer representing the interpolation scheme to be used
                d) for each energy and representation weight point total, the following data is extracted
                    i. [x]: the energy of the incoming neutron {eV}
                    ii. [y]: the representation weight, aka the probability of this representation being used
                e) [nDistFunc]: the number of interpolation schemes used to describe the relationship between out-going hadron energy and the probability of the hadron leaving with that energy. there is one interpolation scheme per incoming neutron kinetic energy so this is also the number of incoming neutron kinetic energy points
                f) [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and the probability of a hadron being ejected with a particular energy.
                g) For each range nRanges, the following data is extracted
                    i. [range]: the end point of the range
                    ii. [it]: an integer representing the interpolation scheme to be used
                h) For each incoming neutron kinetic energy point nDistFunc, the following data is extracted
                    i. [currentEnergy]: the incoming neutron kinetic energy {eV}
                    ii. [total]: the number of outgoing hadron energy and probability of occurrence points
                    iii. [nRanges]: defines the number of interpolation ranges for the interpolation between

outgoing hadron energy and the probability of occurrence.

iv. For each range nRanges, the following data is extracted
[range]: the end point of the range
[it]: an integer representing the interpolation scheme to be used

v. For each outgoing hadron energy point total, the following data is extracted
[x]: the out-going hadron energy {eV}
[y]: the probability of occurrence

3. if theRepresentationType=5
   a) For neatness the following is organised into three parts
   b) theFractionalProb
      i. [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
      ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.
      iii. For each range nRanges, the following data is extracted
      [range]: the end point of the range
      [it]: an integer representing the interpolation scheme to be used
      iv. for each energy and representation weight point total, the following data is extracted
      [x]: the energy of the incoming neutron {eV}
      [y]: the representation weight, aka the probability of this representation being used
   c) theThetaDist
      i. [total]: the total number of incoming neutron kinetic energy and average outgoing hadron energy points
      ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and average out-going neutron energy.
      iii. For each range nRanges, the following data is extracted
      [range]: the end point of the range
      [it]: an integer representing the interpolation scheme to be used
      iv. for each incoming neutron kinetic energy and maximum out-going hadron energy point total, the following data is extracted
      [x]: the incoming neutron kinetic energy {eV}
      [y]: the average out-going hadron energy
   d) theXDist
      i. [total]: the number of multiplier and weight points. The weights are the probability of the multiplier being randomly selected. When an outgoing hadron energy is called for, the average outgoing hadron energy is interpolated from the ThetaDist data and a randomly selected multiplier is applied to it.
      ii. [nRanges]: defines the number of interpolation ranges for the interpolation between multiplier and weight points.
      iii. For each range nRanges, the following data is extracted
      [range]: the end point of the range
      [it]: an integer representing the interpolation scheme to be used
      iv. for each energy and representation weight point total, the following data is extracted
      [x]: the multiplier
      [y]: the weight
4. if theRepresentationType=7,9
   a) theFractionalProb
      i. [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
      ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.
      iii. For each range nRanges, the following data is extracted
      [range]: the end point of the range

[it]: an integer representing the interpolation scheme to be used
iv. for each energy and representation weight point total, the following data is extracted
[x]: the energy of the incoming neutron {eV}
[y]: the representation weight, aka the probability of this representation being used

b) theThetaDist
i. [total]: the total number of incoming neutron kinetic energy and average outgoing hadron energy points
ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and average out-going neutron energy.
iii. For each range nRanges, the following data is extracted
[range]: the end point of the range
[it]: an integer representing the interpolation scheme to be used
iv. for each incoming neutron kinetic energy and maximum out-going hadron energy point total, the following data is extracted
[x]: the incoming neutron kinetic energy {eV}
[y]: the average out-going hadron energy {eV}

5. if theRepresentationType=11
a) theFractionalProb
i. [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.
iii. For each range nRanges, the following data is extracted
[range]: the end point of the range
[it]: an integer representing the interpolation scheme to be used
iv. for each energy and representation weight point total, the following data is extracted
[x]: the energy of the incoming neutron {eV}
[y]: the representation weight, aka the probability of this representation being used

b) theApar
i. [total]: the total number of incoming neutron kinetic energy and parameter A of the Watt Spectrum points
ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and parameter A of the Watt Spectrum.
iii. For each range nRanges, the following data is extracted
[range]: the end point of the range
[it]: an integer representing the interpolation scheme to be used
iv. for each incoming neutron kinetic energy and parameter A of the Watt Spectrum point total, the following data is extracted
[x]: the incoming neutron kinetic energy {eV}
[y]: the parameter A of the Watt Spectrum

c) theBpar
i. [total]: the total number of incoming neutron kinetic energy and parameter B of the Watt Spectrum points
ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and parameter B of the Watt Spectrum.
iii. For each range nRanges, the following data is extracted
[range]: the end point of the range
[it]: an integer representing the interpolation scheme to be used
iv. for each incoming neutron kinetic energy and parameter A of the Watt Spectrum point total, the following data is extracted
[x]: the incoming neutron kinetic energy {eV}
[y]: the parameter B of the Watt Spectrum

6. if theRepresentationType=12
a) theFractionalProb
i. [total]: the number of energy and representation weight points in this interpolation

scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.

ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.

iii. For each range nRanges, the following data is extracted
[range]: the end point of the range
[it]: an integer representing the interpolation scheme to be used

iv. for each energy and representation weight point total, the following data is extracted
[x]: the energy of the incoming neutron {eV}
[y]: the representation weight, aka the probability of this representation being used

b) [theAvarageKineticPerNucleonForLightFragments]: the average kinetic energy of a light fragment per nucleon in the fragment {eV}

c) [theAvarageKineticPerNucleonForHeavyFragments]: the average kinetic energy of a heavy fragment per nucleon in the fragment {eV}

d) theMaxTemp

i. [total]: the total number of incoming neutron kinetic energy and the max fission fragment temperature points

ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and max fission fragment temperature.

iii. For each range nRanges, the following data is extracted
[range]: the end point of the range
[it]: an integer representing the interpolation scheme to be used

iv. for each incoming neutron kinetic energy and parameter A of the Watt Spectrum point total, the following data is extracted
[x]: the incoming neutron kinetic energy {eV}
[y]: the max fission fragment temperature

f) if infoType=4

1. if dataType=1

A) theFinalStateNeutrons.InitPrompt

B) this section defines the relationship between incoming neutron kinetic energy and outgoing prompt neutron multiplicity

i. [targetMass]: the mass of the isotope {/(mass of a neutron)}

ii. [iflag]: specifies whether the prompt neutron multiplicity will be constant or interpolated from the incoming neutron kinetic energy

iii. if iflag!=2

1. [theSpontPrompt]: the multiplicity (i.e. number) of prompt neutrons that shall be created for all incoming neutron energies, for this isotope.

iv. if iflag=2

1. [total]: the total number of incoming neutron kinetic energy and outgoing prompt neutron multiplicity points
2. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and outgoing prompt neutron multiplicity.
3. For each range nRanges, the following data is extracted
   a) [range]: the end point of the range
   b) [it]: an integer representing the interpolation scheme to be used
4. for each incoming neutron kinetic energy and outgoing prompt neutron multiplicity point total, the following data is extracted
   a) [x]: the incoming neutron kinetic energy {eV}
   b) [y]: the outgoing prompt neutron multiplicity

g) if infoType=5

1. if dataType=1

A) theEnergyRelease.Init

i. [dummy]: not used for anything

ii. [fragmentKinetic]: the kinetic energy of the produced fission fragments {eV}

iii. [promptNeutronKinetic]: not used for anything

iv. [delayedNeutronKinetic]: not used for anything

v. [promptGammaEnergy]: not used for anything
vi. [delayedGammaEnergy]: not used for anything
vii. [delayedBetaEnergy]: not used for anything
viii. [neutrinoEnergy]: not used for anything
ix. [reducedTotalEnergy]: not used for anything
x. [totalEnergy]: not used for anything

2. FC/, SC/, TC/, and LC/
   A) Each of the directories are different fission reactions which determine the properties of the out-going prompt neutrons. The directory names corespond to First Chance fission MT=19, Second Chance fission MT=20, Third Chance fission MT=21, and Last Chance fission MT=38 respectively
   B) [dummy]: the MT number, not used for anything
   C) [dummy]: not used for anything
   D) [total]: the number of incoming neutron kinetic energy and cross-section (probability of the reaction occurring) points
   E) for each incoming neutron kinetic energy and cross-section point total, the following data is extracted
      a) [x]: incoming neutron kinetic energy {eV}
      b) [y]: cross-section (probability of this model being used)
   F) [dummy]: not used for anything
   G) [dummy]: not used for anything
   H) theAngularDistribution.Init
      a) The data extracted in this section defines the probability that a hadron with a particular angle will be produced by an incoming neutron with a particular energy if this inelastic scattering reaction occurs
      b) [theAngularDistributionType]: determines how the angular distribution of the produced hadrons are to be represented and subsequently the format of the data to be extracted
      c) [targetMass]: the mass of the isotope {/(mass of a neutron)}
      d) [frameFlag]: the reference frame that the data was measured with respect to (1=LAB, 2=CMS)
      e) if theAngularDistributionType = 0
         1. the angular distribution of the produced hadrons is to be represented by an isotropic distribution which means that no further data is necessary
      f) if theAngularDistributionType = 1
         1. The angular distribution of the produced hadrons (remember the incoming is absorbed) is to be represented by Legendre polynomial sets which represent the probability angular relationship. There is a set of Legendre polynomials for each incoming neutron kinetic energy and each of the polynomials in a set are centered around a particular exit angle (meaning that the closer the angle is to the angle the Legendre polynomial is centered on, the better the approximation of that polynomial becomes). The coefficients of the Legendre polynomials are dependent on energy of the incoming neutrons and the angle of the exiting hadron.
         2. [nEnergy]: the number of incoming neutron kinetic energy points and also the number of Legendre polynomial sets
         3. [nRanges]: defines the number of interpolation ranges for the interpolation of between the Legendre polynomial sets.
         4. For each range nRanges, the following data is extracted
            A) [range]: the end point of the range
            B) [it]: an integer representing the interpolation scheme to be used
         5. for each Legendre polynomial set nEnergy, the following data is extracted
            A) [temp]: the temperature of the isotope {K}
            B) [energy]: the energy of the incoming neutron {eV}
            C) [tempdep]: not used for anything
            D) [nLegendre]: the number of Legendre polynomials in the set and also the number of exit angle probability points
            E) for each Legendre polynomial nLegendre, the following data is extracted
               i. [coeff]: the coefficient of the Legendre polynomial
               ii. note: the angle that the polynomial is centered around is based off the indices of the polynomial in the set
      g) if theAngularDistributionType = 2
         1. The angular distribution is represented by a 3D table of incoming neutron kinetic energy, exit angle and probability
         2. [nEnergy]: the number of incoming neutron kinetic energy points

3. [nRanges]: defines the number of interpolation ranges for the interpolation of between the incoming neutron kinetic energy points.
4. For each range nRanges, the following data is extracted
   A) [range]: the end point of the range
   B) [it]: an integer representing the interpolation scheme to be used
5. for each incoming neutron kinetic energy point nEnergy, the following data is extracted
   A) [temp]: the temperature of the isotope that the data was taken at
   B) [energy]: the energy of the incoming neutron kinetic energy {eV}
   C) [tempdep]: not used for anything
   D) [neg]: the number of exit angle points for this incoming neutron kinetic energy
   E) [nRanges]: defines the number of interpolation ranges for the interpolation of between the exit angle points.
   F) For each range nRanges, the following data is extracted
      i. [range]: the end point of the range
      ii. [it]: an integer representing the interpolation scheme to be used
   G) for each exit angle point neg, the following data is extracted
      i. [eg]: the exit angle of the hadron {eV}
      ii. [pg]: the probability of this hadron being created

I) [dummy]: not used for anything
J) [dummy]: not used for anything
K) theEnergyDistribution.Init
   a) [dummy]: does nothing
   b) [theNumberOfPartials]: the number of partials that will make up interpolation scheme between incoming neutron kinetic energy and outgoing hadron energy
   c) for each partial theNumberOfPartials, the following data will be extracted
      1. [theRepresentationType]: Specifies how the data has been formatted
      2. if theRepresentationType=1
         A) [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
         B) [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.
         C) For each range nRanges, the following data is extracted
            i. [range]: the end point of the range
            ii. [it]: an integer representing the interpolation scheme to be used
         D) for each energy and representation weight point total, the following data is extracted
            i. [x]: the energy of the incoming neutron {eV}
            ii. [y]: the representation weight, aka the probability of this representation being used
         E) [nDistFunc]: the number of interpolation schemes used to describe the relationship between out-going hadron energy and the probability of the hadron leaving with that energy. there is one interpolation scheme per incoming neutron kinetic energy so this is also the number of incoming neutron kinetic energy points
         F) [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and the energy probability distribution of the ejected hadron.
         G) For each range nRanges, the following data is extracted
            i. [range]: the end point of the range
            ii. [it]: an integer representing the interpolation scheme to be used
         H) For each incoming neutron kinetic energy point nDistFunc, the following data is extracted
            i. [currentEnergy]: the incoming neutron kinetic energy {eV}
            ii. [total]: the number of outgoing hadron energy and probability of occurrence points
            iii. [nRanges]: defines the number of interpolation ranges for the interpolation between outgoing hadron energy and the probability of occurrence.
            iv. For each range nRanges, the following data is extracted
               1. [range]: the end point of the range
               2. [it]: an integer representing the interpolation scheme to be used
            v. For each outgoing hadron energy point total, the following data is extracted

1. [x]: the out-going hadron energy {eV}
2. [y]: the probability of occurrence

3. if theRepresentationType=5
   A) For neatness the following is organised into three parts
   B) theFractionalProb
      i. [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
      ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.
      iii. For each range nRanges, the following data is extracted
         1. [range]: the end point of the range
         2. [it]: an integer representing the interpolation scheme to be used
      iv. for each energy and representation weight point total, the following data is extracted
         1. [x]: the energy of the incoming neutron {eV}
         2. [y]: the representation weight, aka the probability of this representation being used
   C) theThetaDist
      i. [total]: the total number of incoming neutron kinetic energy and average outgoing hadron energy points
      ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and average out-going neutron energy.
      iii. For each range nRanges, the following data is extracted
         1. [range]: the end point of the range
         2. [it]: an integer representing the interpolation scheme to be used
      iv. for each incoming neutron kinetic energy and maximum out-going hadron energy point total, the following data is extracted
         1. [x]: the incoming neutron kinetic energy {eV}
         2. [y]: the average out-going hadron energy
   D) theXDist
      i. [total]: the number of multiplier and weight points. The weights are the probability of the multiplier being randomly selected. When an outgoing hadron energy is called for, the average outgoing hadron energy is interpolated from the ThetaDist data and a randomly selected multiplier is applied to it.
      ii. [nRanges]: defines the number of interpolation ranges for the interpolation between multiplier and weight points.
      iii. For each range nRanges, the following data is extracted
         1. [range]: the end point of the range
         2. [it]: an integer representing the interpolation scheme to be used
      iv. for each energy and representation weight point total, the following data is extracted
         1. [x]: the multiplier
         2. [y]: the weight

4. if theRepresentationType=7,9
   A) theFractionalProb
      i. [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
      ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.
      iii. For each range nRanges, the following data is extracted
         1. [range]: the end point of the range
         2. [it]: an integer representing the interpolation scheme to be used
      iv. for each energy and representation weight point total, the following data is extracted
         1. [x]: the energy of the incoming neutron {eV}
         2. [y]: the representation weight, aka the probability of this representation being used
   B) theThetaDist
      i. [total]: the total number of incoming neutron kinetic energy and average outgoing hadron energy

points
   ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and average out-going neutron energy.
   iii. For each range nRanges, the following data is extracted
      1. [range]: the end point of the range
      2. [it]: an integer representing the interpolation scheme to be used
   iv. for each incoming neutron kinetic energy and maximum out-going hadron energy point total, the following data is extracted
      1. [x]: the incoming neutron kinetic energy {eV}
      2. [y]: the average out-going hadron energy {eV}

5. if theRepresentationType=11
   A) theFractionalProb
      i. [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
      ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.
      iii. For each range nRanges, the following data is extracted
         1. [range]: the end point of the range
         2. [it]: an integer representing the interpolation scheme to be used
      iv. for each energy and representation weight point total, the following data is extracted
         1. [x]: the energy of the incoming neutron {eV}
         2. [y]: the representation weight, aka the probability of this representation being used
   B) theApar
      i. [total]: the total number of incoming neutron kinetic energy and parameter A of the Watt Spectrum points
      ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and parameter A of the Watt Spectrum.
      iii. For each range nRanges, the following data is extracted
         1. [range]: the end point of the range
         2. [it]: an integer representing the interpolation scheme to be used
      iv. for each incoming neutron kinetic energy and parameter A of the Watt Spectrum point total, the following data is extracted
         1. [x]: the incoming neutron kinetic energy {eV}
         2. [y]: the parameter A of the Watt Spectrum
   C) theBpar
      i. [total]: the total number of incoming neutron kinetic energy and parameter B of the Watt Spectrum points
      ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and parameter B of the Watt Spectrum.
      iii. For each range nRanges, the following data is extracted
         1. [range]: the end point of the range
         2. [it]: an integer representing the interpolation scheme to be used
      iv. for each incoming neutron kinetic energy and parameter A of the Watt Spectrum point total, the following data is extracted
         1. [x]: the incoming neutron kinetic energy {eV}
         2. [y]: the parameter B of the Watt Spectrum
6. if theRepresentationType=12
   A) theFractionalProb
      i. [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
      ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.
      iii. For each range nRanges, the following data is extracted

1. [range]: the end point of the range
2. [it]: an integer representing the interpolation scheme to be used

iv. for each energy and representation weight point total, the following data is extracted
1. [x]: the energy of the incoming neutron {eV}
2. [y]: the representation weight, aka the probability of this representation being used

B) [theAvarageKineticPerNucleonForLightFragments]: the average kinetic energy of a light fragment per nucleon in the fragment {eV}

C) [theAvarageKineticPerNucleonForHeavyFragments]: the average kinetic energy of a heavy fragment per nucleon in the fragment {eV}

D) theMaxTemp

i. [total]: the total number of incoming neutron kinetic energy and the max fission fragment temperature points

ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and max fission fragment temperature.

iii. For each range nRanges, the following data is extracted
1. [range]: the end point of the range
2. [it]: an integer representing the interpolation scheme to be used

iv. for each incoming neutron kinetic energy and parameter A of the Watt Spectrum point total, the following data is extracted
1. [x]: the incoming neutron kinetic energy {eV}
2. [y]: the max fission fragment temperature {k}

3. FF/

A) while there is data left

a) [iMT]: the ENDF MT number, identifies the reaction type, only reaction type MT=454 is used

b) [iMF]: not used for anything, the ENDF MF number, identifies what type of data is contained in this section of the file (could be photon yield data or outgoing neutron energy distribution, etc.)

c) [dummy]: does nothing

d) [imax]: the number of incoming neutron kinetic energy points

e) for each incoming neutron kinetic energy point imax, the following data is extracted
1. [Ei]: the kinetic energy of the incoming neutron {eV}
2. [jmax]: The number of possible fission fragments that can be produced by this reaction at this incoming neutron kinetic energy
3. [ip]: not used for anything, the interpolation scheme for interpolating between incoming neutron kinetic energy points
4. for jmax, the following data is extracted

A) [FSP]: the Z value of the fission fragment multiplied by 1000 (FSP stands for Final State Product)

B) [mFSP]: the decimal part of the number that identifies what state the fission product is in (ground, 1st excited, etc.)

C) [Y]: the yield (in this case the probability) of this fission fragment being created for this reaction and incoming neutron kinetic energy

Inelastic Scattering

There are 36 different directories containing the final state data, each directory represents a different reaction. All of the different reactions use either G4NeutronHPInelasticBaseFS.Init() or G4NeutronHPInelasticCompFS.Init(), to extract their data

Directories: F02, F03, F04, F05, F06, F07, F08, F09, F10, F11, F12, F13, F14, F15, F16, F17, F18, F19, F20, F21, F22, F28, F29, F30, F31, F32, F33, F34, F35, F36, use G4NeutronHPInelasticBaseFS::Init()

G4NeutronHPInelasticBaseFS::Init ()

[infoType]: is an integer used to check if the file has any data left to extract, the extraction operation returns true if it is successful in extracting the integer infoType

while the extraction of infoType returns true, the following data is extracted

1. [dataType]: determines what type of data is to be extracted
2. if dummy=INT_MAX (this is true for at least the first pass)

A) [dummy]: is extracted twice, does nothing but hold a place

3. if dataType=3

the data extracted in this section describes the probability of this specific inelastic scattering reaction occurring using a simple interpolation scheme between energy and cross-section
[total]: the number of energy cross-section points
for each energy cross-section point total, the following data is extracted
i. [x]: the energy that the cross-section was measured at in {eV}
ii. [y]: the cross-section value in {barns}

4. if dataType=4
   A) The data extracted in this section defines the probability that a hadron with a particular angle will be produced by an incoming neutron with a particular energy if this inelastic scattering reaction occurs
   B) [theAngularDistributionType]: determines how the angular distribution of the produced hadrons are to be represented and subsequently the format of the data to be extracted
   C) [targetMass]: the mass of the isotope {/(mass of a neutron)}
   D) [frameFlag]: the reference frame that the data was measured with respect to (1=LAB, 2=CMS)
   E) if theAngularDistributionType = 0
      i. the angular distribution of the produced hadrons is to be represented by an isotropic distribution which means that no further data is necessary
   F) if theAngularDistributionType = 1
      i. The angular distribution of the produced hadrons (remember the incoming is absorbed) is to be represented by Legendre polynomial sets which represent the probability angular relationship. There is a set of Legendre polynomials for each incoming neutron kinetic energy and each of the polynomials in a set are centered around a particular exit angle (meaning that the closer the angle is to the angle the Legendre polynomial is centered on, the better the approximation of that polynomial becomes). The coefficients of the Legendre polynomials are dependent on energy of the incoming neutrons and the angle of the exiting hadron.
      ii. [nEnergy]: the number of incoming neutron kinetic energy points and also the number of Legendre polynomial sets
      iii. [nRanges]: defines the number of interpolation ranges for the interpolation of between the incoming neutron kinetic energy and the Legendre polynomial sets.
      iv. For each range nRanges, the following data is extracted
         a) [range]: the end point of the range
         b) [it]: an integer representing the interpolation scheme to be used
      v. for each Legendre polynomial set nEnergy, the following data is extracted
         a) [temp]: the temperature of the isotope
         b) [energy]: the energy of the incoming neutron {eV}
         c) [tempdep]: not used for anything
         d) [nLegendre]: the number of Legendre polynomials in the set and also the number of exit angle probability points
         e) for each Legendre polynomial nLegendre, the following data is extracted
            1. [coeff]: the coefficient of the Legendre polynomial
            2. note: the angle that the polynomial is centered around is based off the indices of the polynomial in the set
   G) if theAngularDistributionType = 2
      i. The angular distribution is represented by a 3D table of incoming neutron kinetic energy, exit angle and probability
      ii. [nEnergy]: the number of incoming neutron kinetic energy points
      iii. [nRanges]: defines the number of interpolation ranges for the interpolation of between the incoming neutron kinetic energy points.
      iv. For each range nRanges, the following data is extracted
         a) [range]: the end point of the range
         b) [it]: an integer representing the interpolation scheme to be used
      v. for each incoming neutron kinetic energy point nEnergy, the following data is extracted
         a) [temp]: the temperature of the isotope that the data was taken at
         b) [energy]: the energy of the incoming neutron kinetic energy
         c) [tempdep]: not used for anything
         d) [neg]: the number of exit angle points for this incoming neutron kinetic energy
         e) [nRanges]: defines the number of interpolation ranges for the interpolation of between the exit angle points.
         f) For each range nRanges, the following data is extracted

1. [range]: the end point of the range
2. [it]: an integer representing the interpolation scheme to be used

g) for each exit angle point neg, the following data is extracted
1. [eg]: the exit angle of the hadron
2. [pg]: the probability of this hadron being created

5. if dataType=5

A) theEnergyDistribution->Init()

B) The data collected in this section defines the energy distribution of the produced hadrons using various models. Each model contains an incoming neutron kinetic energy and model weight (the model accuracy at a particular incoming neutron kinetic energy) interpolation scheme which allows the process to stochastically select a model to be used for a particular incoming neutron

C) [dummy]: does nothing

D) [theNumberOfPartials]: the number of partials that will make up interpolation scheme between incoming neutron kinetic energy and outgoing hadron energy

E) for each partial theNumberOfPartials, the following data will be extracted

i. [theRepresentationType]: Specifies how the data has been formatted

ii. if theRepresentationType=1

a) [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.

b) [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.

c) For each range nRanges, the following data is extracted
1. [range]: the end point of the range
2. [it]: an integer representing the interpolation scheme to be used

d) for each energy and representation weight point total, the following data is extracted
1. [x]: the energy of the incoming neutron {eV}
2. [y]: the representation weight, aka the probability of this representation being used

e) [nDistFunc]: the number of interpolation schemes used to describe the relationship between out-going hadron energy and the probability of the hadron leaving with that energy. there is one interpolation scheme per incoming neutron kinetic energy so this is also the number of incoming neutron kinetic energy points

f) [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and the energy probability of the ejected hadron.

g) For each range nRanges, the following data is extracted
1. [range]: the end point of the range
2. [it]: an integer representing the interpolation scheme to be used

h) For each incoming neutron kinetic energy point nDistFunc, the following data is extracted
1. [currentEnergy]: the incoming neutron kinetic energy {eV}
2. [total]: the number of outgoing hadron energy and probability of occurrence points
3. [nRanges]: defines the number of interpolation ranges for the interpolation between outgoing hadron energy and the probability of occurrence.
4. For each range nRanges, the following data is extracted
i. [range]: the end point of the range
ii. [it]: an integer representing the interpolation scheme to be used
5. For each outgoing hadron energy point total, the following data is extracted
i. [x]: the out-going hadron energy {eV}
ii. [y]: the probability of occurrence

iii. if theRepresentationType=5

a) For neatness the following is organised into three parts

b) theFractionalProb
1. [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
2. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming

neutron kinetic energy and representation weight points.
3. For each range nRanges, the following data is extracted
   i. [range]: the end point of the range
   ii. [it]: an integer representing the interpolation scheme to be used
4. for each energy and representation weight point total, the following data is extracted
   i. [x]: the energy of the incoming neutron {eV}
   ii. [y]: the representation weight, aka the probability of this representation being used

c) theThetaDist
1. [total]: the total number of incoming neutron kinetic energy and average outgoing hadron energy points
2. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and average out-going neutron energy.
3. For each range nRanges, the following data is extracted
   i. [range]: the end point of the range
   ii. [it]: an integer representing the interpolation scheme to be used
4. for each incoming neutron kinetic energy and maximum out-going hadron energy point total, the following data is extracted
   i. [x]: the incoming neutron kinetic energy {eV}
   ii. [y]: the average out-going hadron energy

d) theXDist
1. [total]: the number of multiplier and weight points. The weights are the probability of the multiplier being randomly selected. When an outgoing hadron energy is called for, the average outgoing hadron energy is interpolated from the ThetaDist data and a randomly selected multiplier is applied to it.
2. [nRanges]: defines the number of interpolation ranges for the interpolation between multiplier and weight points.
3. For each range nRanges, the following data is extracted
   i. [range]: the end point of the range
   ii. [it]: an integer representing the interpolation scheme to be used
4. for each energy and representation weight point total, the following data is extracted
   i. [x]: the multiplier
   ii. [y]: the weight

iv. if theRepresentationType=7,9

a) theFractionalProb
1. [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
2. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.
3. For each range nRanges, the following data is extracted
   i. [range]: the end point of the range
   ii. [it]: an integer representing the interpolation scheme to be used
4. for each energy and representation weight point total, the following data is extracted
   i. [x]: the energy of the incoming neutron {eV}
   ii. [y]: the representation weight, aka the probability of this representation being used

b) theThetaDist
1. [total]: the total number of incoming neutron kinetic energy and average outgoing hadron energy points
2. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and average out-going neutron energy.
3. For each range nRanges, the following data is extracted
   i. [range]: the end point of the range
   ii. [it]: an integer representing the interpolation scheme to be used
4. for each incoming neutron kinetic energy and maximum out-going hadron energy point total, the following data is extracted
   i. [x]: the incoming neutron kinetic energy {eV}
   ii. [y]: the average out-going hadron energy {eV}

v. if theRepresentationType=11
   a) theFractionalProb
      1. [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
      2. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.
      3. For each range nRanges, the following data is extracted
         i. [range]: the end point of the range
         ii. [it]: an integer representing the interpolation scheme to be used
      4. for each energy and representation weight point total, the following data is extracted
         i. [x]: the energy of the incoming neutron {eV}
         ii. [y]: the representation weight, aka the probability of this representation being used
   b) theApar
      1. [total]: the total number of incoming neutron kinetic energy and parameter A of the Watt Spectrum points
      2. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and parameter A of the Watt Spectrum.
      3. For each range nRanges, the following data is extracted
         i. [range]: the end point of the range
         ii. [it]: an integer representing the interpolation scheme to be used
      4. for each incoming neutron kinetic energy and parameter A of the Watt Spectrum point total, the following data is extracted
         i. [x]: the incoming neutron kinetic energy {eV}
         ii. [y]: the parameter A of the Watt Spectrum
   c) theBpar
      1. [total]: the total number of incoming neutron kinetic energy and parameter B of the Watt Spectrum points
      2. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and parameter B of the Watt Spectrum.
      3. For each range nRanges, the following data is extracted
         i. [range]: the end point of the range
         ii. [it]: an integer representing the interpolation scheme to be used
      4. for each incoming neutron kinetic energy and parameter A of the Watt Spectrum point total, the following data is extracted
         i. [x]: the incoming neutron kinetic energy {eV}
         ii. [y]: the parameter B of the Watt Spectrum
vi. if theRepresentationType=12
   a) theFractionalProb
      1. [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
      2. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.
      3. For each range nRanges, the following data is extracted
         i. [range]: the end point of the range
         ii. [it]: an integer representing the interpolation scheme to be used
      4. for each energy and representation weight point total, the following data is extracted
         i. [x]: the energy of the incoming neutron {eV}
         ii. [y]: the representation weight, aka the probability of this representation being used
   b) [theAvarageKineticPerNucleonForLightFragments]: the average kinetic energy of a light fragment per nucleon in the fragment {eV}
   c) [theAvarageKineticPerNucleonForHeavyFragments]: the average kinetic energy of a heavy fragment per nucleon in the fragment {eV}
   d) theMaxTemp

1. [total]: the total number of incoming neutron kinetic energy and the max fission fragment temperature points
2. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and max fission fragment temperature.
3. For each range nRanges, the following data is extracted
   i. [range]: the end point of the range
   ii. [it]: an integer representing the interpolation scheme to be used
4. for each incoming neutron kinetic energy and parameter A of the Watt Spectrum point total, the following data is extracted
   i. [x]: the incoming neutron kinetic energy {eV}
   ii. [y]: the max fission fragment temperature

6. if dataType=6
   A) The data in this section describes the energy-angular distribution of the outgoing neutrons
   B) [targetMass]: the mass of the isotope {/(mass of a neutron)}
   C) [frameFlag]: the relativistic frame that the data is taken with respect to
   D) [nProducts]: the number of states that the nucleus can go into after the absorption of the neutron
   E) for each states of the nucleus [nProducts], the following data is extracted
      i. [theMassCode]: the 5 digit ZZAAA code for the out going particle
      ii. [theMass]: not used for anything
      iii. [theIsomerFlag]: not used for anything
      iv. [theDistLaw]: specifies the distribution of the neutrons, i.e. isotropic, continuous angular-energy dist, discrete 2 body scattering
      v. [theGroundStateQValue]: not used for anything, the energy difference between is excited state and the ground state, i.e. the theoretical Q-value of the reaction {eV}
      vi. [theActualStateQValue]: the energy difference between the excited state after absorption and the ground state before absorption, i.e. the actual Q-value of the reaction, should be very close to the theoretical {eV}
      vii. [total]: the total number of incoming neutron kinetic energy and out-going neutron multiplicity (the number of neutrons released) points.
      viii. [nRanges]: defines the number of interpolation ranges for the interpolation between the incoming neutron kinetic energy and out-going neutron multiplicity.
      ix. For each range [nRanges], the following data is extracted
         a) [range]: the end point of the range
         b) [it]: an integer representing the interpolation scheme to be used
      x. for each out-going neutron multiplicity point [total], the following data is extracted
         a) [x]: the incoming neutron kinetic energy {eV}
         b) [y]: the out-going neutron multiplicity (ie the number of neutrons produced if this reaction occurs)
      xi. if theDistLaw=0,3
         a) An isotropic neutron angular distribution is used
         b) No data is collected from the file, instead uniform random distributions are used to sample the angles
      xii. if theDistLaw=1
         a) A continuous energy and angular distribution is used
         b) [theTargetCode]: is the ZZAAA integer that specifies the nucleus
         c) [theAngularRep]: is an integer that specifies how the angular distribution is to be sampled
         d) [theInterpolation]: specifies the interpolation scheme to be used for the interpolation between probability of the neutron being created with a certain energy and angle, and the incoming neutron kinetic energy
         e) [nEnergy]: the number of incoming neutron kinetic energy points
         f) [nRanges]: defines the number of interpolation ranges for the interpolation of between the out-going energy and angle sets.
         g) For each range [nRanges], the following data is extracted
            1. [range]: the end point of the range
            2. [it]: an integer representing the interpolation scheme to be used
         h) for each incoming neutron kinetic energy point [nEnergy], the following data is extracted
            1. [theEnergy]: the incoming neutron kinetic energy {eV}
            2. [nEnergies]: the number of outgoing neutron energy points
            3. [nDiscreteEnergies]: the number of neutron energy points that have discrete energies i.e. the number of points that are not part of the continuous neutron energy dist (the neutron can only occur at the energy point, there is no interpolation in-between the points), thus nDiscreteEnergies<= nEnergies,

and nEnergies- nDiscreteEnergies is the number of neutron energy points that are part of the continuous energy dist.

4. [nAngularParameters]: the number angle probability points for each neutron energy point
5. for each neutron energy point [nEnergies], the following data is extracted
   i. [sEnergy]: the out-going neutron energy {eV}
   ii. for each angle probability point nAngularParameters, the following data is extracted
      a) [y]: the probability of the occurrence of the neutron (the neutron meaning at the specified isotope state, incoming neutron kinetic energy, neutron energy, and angle). Note the angle corresponding to the probability is based off the probability array index, i.e. each array index in the probability array is a set angle apart.

xiii. if theDistLaw=2,4

a) a discrete 2 body recoil distribution is used, this type of distribution uses a set of Legendre polynomials to describe the angular dependence of the probability of a neutron occurring for each neutron energy, there are multiple neutron energies for each incoming neutron kinetic energy
b) [nEnergy]: the number of neutron energy points
c) [nRanges]: defines the number of interpolation ranges for the interpolation of between the Legendre polynomial sets.
d) For each range [nRanges], the following data is extracted
   1. [range]: the end point of the range
   2. [it]: an integer representing the interpolation scheme to be used
e) for each neutron energy point [nEnergy], the following data is collected
   1. [energy]: the incoming neutron kinetic energy {eV}
   2. [aRep]: the interpolation scheme between the Legendre coefficients
   3. [nCoeff]: the number of Legendre polynomial coefficients
   4. if aRep>0
      i. then the total number of Legendre coefficients is 2xnCoeff
      ii. for the total number of Legendre coefficients, the following data is extracted
         a) [y]: is Legendre polynomial coefficient
   5. if aRep=0 then
      i. then the total number of Legendre coefficients is nCoeff
      ii. for the total number of Legendre coefficients, the following data is extracted
         a) [y]: is Legendre polynomial coefficient

xiv. if theDistLaw=5

a) does nothing, has not been coded yet

xv. if theDistLaw=6

a) an n-body phase space distribution is used
b) [theTotalMass]: the mass of the isotope divided by the mass of a neutron {MeV}
c) [theTotalCount]: the number of particles involved in the reaction

xvi. if theDistLaw=7

a) an experimental parameterized energy and angular model
b) [nEnergies]: the number of incoming neutron energies
c) [nRanges]: defines the number of interpolation ranges for the interpolation of between incoming neutron kinetic energy and the probability of the neutron being shot in at a specific angle.
d) For each range [nRanges], the following data is extracted
   1. [range]: the end point of the range
   2. [it]: an integer representing the interpolation scheme to be used
e) for each incoming neutron kinetic energy [nEnergies], the following data is collected
   1. [theEnergies]: the incoming neutron kinetic energy {eV}
   2. [nCosTh]: the number of angle points for this incoming neutron kinetic energy
   3. [nRanges]: defines the number of interpolation ranges for the interpolation of between the neutron angular direction and the probability of the neutron having a specific energy.
   4. For each range [nRanges], the following data is extracted
      i. [range]: the end point of the range
      ii. [it]: an integer representing the interpolation scheme to be used
   5. for each angle point [nCosTh], the following data is extracted
      i. [label]: this is the cosine of the angle
      ii. [total]: the number of neutron energy points

iii. [nRanges]: defines the number of interpolation ranges for the interpolation of between the neutron energy and the probability of the neutron occurring.
iv. For each range nRanges, the following data is extracted
    a) [range]: the end point of the range
    b) [it]: an integer representing the interpolation scheme to be used
v. for each neutron energy point [total], the following information is extracted
    a) [x]: the neutron energy {eV}
    b) [y]: the probability of occurrence

7. if dataType=12
   A) defines the relationship between outgoing photon energy, the probability of the photon occurring and the fraction of photons that will transfer their kinetic energy to an electron
   B) The first number [repFlag] refers to the format that the data is presented in
   C) if repFlag can be extracted
      i. The second number [targetMass] is the mass of the isotope in {/(mass of a neutron)}
      ii. Now depending on the repFlag the rest of the data is extracted in the following ways
      iii. repFlag=1
         a) this format uses a simple interpolation scheme to represent the dependence of photon multiplicity on the incoming neutron kinetic energy, for each possible discrete photon energy
         b) the next number [nDiscrete] is the number of discrete types of photons that can be emitted for this isotope
         c) For each discrete type of photon the following data is collected
            1. [disType]: the photon energy distribution type (continuous [1] or discrete [0]) , determines whether the photon energy will be interpolated or not
            2. [energy]: the energy of the produced photon in {eV}
            3. [total]: the number of incoming neutron kinetic energy points
            4. [nRanges]: defines the number of interpolation ranges (see [1] for explanation of ranges) for the interpolation of incoming neutron kinetic energy and the photon multiplicity.
            5. For each range [nRanges], the following data is extracted
               i. [range]: the end point of the range
               ii. [it]: an integer representing the interpolation scheme to be used
            6. For each incoming neutron kinetic energy point [total], the following data is extracted
               i. [x]: the incoming neutron kinetic energy {eV}
               ii. [y]: the photon multiplicity (i.e. the number of photons produced)
      iv. repFlag=2
         a) This format is a table of photon energy vs the relative probability of the photon being produced at that energy. The effects of different incoming neutron energies on the photon energy are taken care of later in the application of the process to the track
         b) [theInternalConversionFlag]: this number isn't used for anything
         c) [theBaseEnergy]: is the base photon energy used in the creation of photons by the inelastic process {eV}
         d) [theInternalConversionFlag]: specifies whether the photon transition fraction data is present in the file
         e) [nGammaEnergies]: specifies the number of photon energy points that there is data for
         f) For each photon energy point [nGammaEnergies], the following data is collected
            1. if theInternalConversionFlag = 1
               i. [theLevelEnergies]: The energy difference between the isotope before the reaction (ground state) and after the photon has been produced (some excited state) {eV}. Photon energy = theBaseEnergy- theLevelEnergies
               ii. [theTransitionProbabilities]: The probability of a photon with this energy being produced
            2. if theInternalConversionFlag = 2
               i. [theLevelEnergies]: The energy difference between the isotope before the reaction (ground state) and after the photon has been produced (some excited state) {eV}. Photon energy = theBaseEnergy- theLevelEnergies
               ii. [theTransitionProbabilities]: The probability of a photon with this energy being produced
               iii. [thePhotonTransitionFraction]: The probability of an electron with the same KE as the photon being released instead of the photon (due to the absorption of the photon by a valence electron)

8. if thedataType=13
   A) the data collected in this section uses a 3D table to describe the probability of a photon with a particular energy being created for various incoming neutron energies, this information is only used in the case that repFlag=0,

currently this section of the code is not meant to be used, setting repFlag to 0 will cause an error

i. [nDiscrete]: the number of photon energy points
ii. [targetMass]: the mass of the isotope
iii. theTotalXsec
    a) None of the data in this section is used for anything
    b) [total]: the number of incoming neutron kinetic energy and total cross-section points
    c) [nRanges]: the number of ranges
    d) For each range nRanges, the following data is extracted
        1. [range]: the end point of the range
        2. [it]: an integer representing the interpolation scheme to be used
    e) for each incoming neutron kinetic energy and total cross-section point total, the following is extracted
        1. [x]: the incoming neutron kinetic energy {eV}
        2. [y]: the total cross-section {barn}
iv. for each photon energy point nDiscrete, the following data is extracted
    a) [theGammas[i]]: the energy of the photon {eV}
    b) [theShells[i]]: not used for anything {eV}
    c) [isPrimary[i]]: not used for anything
    d) [disType[i]]: specifies whether the distribution is discrete or continuous, (if it is discrete there is no interpolation between points)
    e) thePartialXsec[i]
        1. [total]: the number of incoming neutron kinetic energy and cross-section points
        2. [nRanges]: the number of ranges used to interpolate between the incoming neutron kinetic energy and the cross-section of the reaction producing a photon with energy theGammas[i]
        3. For each range nRanges, the following data is extracted
            i. [range]: the end point of the range
            ii. [it]: an integer representing the interpolation scheme to be used
        4. for each incoming neutron kinetic energy and cross-section point total, the following is extracted
            i. [x]: the incoming neutron kinetic energy {eV}
            ii. [y]: the cross-section {barn}

9. if the dataType =14
    A) Defines the relationship between incoming neutron kinetic energy and the probability of a neutron with a specific angle and energy being produced
    B) The next number [isoFlag] specifies whether the distribution of photons is isotropic or not
    C) if isoFlag=1
        i. No more data is collected, the momentum direction angles of created photons will be selected from a uniformly random distribution.
    D) If isoFlag!=1
        i. if repFlag==2
            a) The code spits out an error since this should not occur
        ii. [tabulationType]: specifies how the photon angular probability is tabulated, either as Legendre polynomials, or a simple interpolation scheme (in both cases the probability of occurrence is a function of the cosine of the angle)
        iii. [nDiscrete2]: is the number of discrete photon energies that can be emitted for this isotope, should be equal to nDiscrete
        iv. [nIso]: is the number of isotropic photons (i.e.: photons where the probability is independent of angle)
        v. for each isotropic photon i.e. [nIso], the following data is collected
            a) [theGammas]: The energy of the released photon {eV}
            b) [theShells]: The binding energy of the neutron for a particular shell, this array is not used for anything {eV}
        vi. for each nonisotropic photon [nDiscrete-nIso], the following data is collected
            a) If tabulationType==1 (i.e. photon angular probability is tabulated as Legendre polynomials which depend on the cosine of the direction angle)
                1. [theGammas]: The energy of the released photon {eV}
                2. [theShells]: The binding energy of the neutron for a particular shell, this array is not used for anything {eV}
                3. [nNeu]: the number of incoming neutron kinetic energy points for this photon
                4. The data in this section is extracted but not used for anything

i. [nRanges]: defines the number of interpolation ranges (see [1] for explanation of ranges) for the interpolation of incoming neutron kinetic energy and the Legendre polynomials that relate probability of occurrence to the scattering angle
ii. For each range [nRanges], the following data is extracted
a) [range]: the end point of the range
b) [it]: an integer representing the interpolation scheme to be used
5. for each incoming neutron kinetic energy point [nNeu], the following data is collected
i. [eNeu]: the energy of the incoming neutron {eV}
ii. [nPoly]: the number of the Legendre polynomials used to describe the relationship between cos(theta) (theta is the scattering angle and the probability of occurrence, for this particular case (the domain of a Legendre polynomial is cos(theta) and the range is the probability)
iii. for each Legendre polynomial nPoly, the following data is collected
a) [coeff]: the coefficient of the Legendre polynomial

b) If tabulationType=2 (i.e. photon angular probability is tabulated as a simple interpolation scheme)
1. [theGammas]: The energy of the released photon {eV}
2. [theShells]: The binding energy of the neutron for a particular shell, this array is not used for anything {eV}
3. [nNeu]: the number of incoming neutron kinetic energy points for this photon
4. for each incoming neutron kinetic energy point [nNeu], the following data is collected
i. [eNeu]: the energy of the incoming neutron {eV}
ii. [nProb]: the number of probability points at this energy
iii. [nRanges]: defines the number of interpolation ranges (see [1] for explanation of ranges) for the interpolation of cosTheta and probability of occurrence
iv. For each range [nRanges], the following data is extracted
a) [range]: the end point of the range
b) [it]: an integer representing the interpolation scheme to be used
v. for each probability point [nProb], the following data is collected
a) [cosTheta]: the cosine of the direction angle of the photon
b) [probDist]: the probability of the photon being released under the previous circumstances

10. if the dataType =15
A) In this section the data that defines the continuous photon energy distributions is collected
B) if disType[i]!=1 for any of the photon energy points extracted in InitMean
i. no more data is collected, the energy of created photons will not be interpolated and will simply be the closest discrete photon energy point
C) else
i. [nPartials]: the number of distinct states that the nuclei will assume after absorbing the neutron
ii. for each distinct states that the nuclei will assume [nPartial], the following data is extracted
a) [dummy]: useless number
b) [total]: number of points for interpolating the probability of this state occurring
c) [nRanges]: defines the number of interpolation ranges (see [1] for explanation of ranges) for the interpolation of incoming neutron kinetic energy and the probability of the nucleus entering this distinct state.
d) For each range [nRanges], the following data is extracted
1. [range]: the end point of the range
2. [it]: an integer representing the interpolation scheme to be used
e) for each interpolation point [total], the following data is extracted
1. [x]: the incoming neutron kinetic energy {eV}
2. [y]: the probability of occurrence of this state at the incoming neutron kinetic energy
f) [nen]: number of incoming neutron kinetic energy points for the interpolation of the photon probability distribution
g) [nRanges]: defines the number of interpolation ranges for the interpolation of incoming neutron kinetic energy and the probability of the photon being created with a specific energy.
h) For each range [nRanges], the following data is extracted
1. [range]: the end point of the range
2. [it]: an integer representing the interpolation scheme to be used
i) For each neutron energy point [nen], the following data is extracted
1. [e]: the incoming neutron kinetic energy {eV}

2. [neg]: the number of photon energy points
3. [nRanges]: defines the number of interpolation ranges for the interpolation of the photon energy and the probability of the photon occurring.
4. For each range [nRanges], the following data is extracted
    i. [range]: the end point of the range
    ii. [it]: an integer representing the interpolation scheme to be used
5. For each photon energy point [neg], the following data is extracted
    i. [eg]: the photon energy {eV}
    ii. [pg]: the probability of occurrence

1. Now the data from the appropriate file in the /Inelastic/Gammas/ directory is extracted to describe the produced photons from the reaction
    1. The data collected in this section defines the energy distribution of the photons released by the radioactive decay of the isotope after inelastic scattering has occurred
    2. While there is data left in the file
        A) [levelEnergy]: the excitation energy of the isotope {keV}
        B) if levelEnergy was successfully extracted
            i. [gammaEnergy]: the energy of the produced photon {keV}
            ii. [probability]: the probability of a photon with energy gammaEnergy being produced when the isotope has levelEnergy excitation energy
2. Directories: F01, F23, F24, F25, F26, F27, use G4NeutronHPInelasticCompFS.Init()
3. G4NeutronHPInelasticCompFS.Init()
    1. follows the same format as G4NeutronHPInelasticBaseFS.Init() except for the beginning
    2. [infoType]: is an integer used to check if the file has any data left to extract, the extraction operation returns true if it is successful in extracting the integer infoType
    3. while the extraction of infoType returns true, following data is extracted
        A) [dataType]: determines what type of data is to be extracted
        B) [sfType]: is a mis-spelling of fsType, is the MT num for the reaction and is also used to determine at what index will the extracted data be stored for the relevant arrays
        C) [dummy]: does nothing but hold a place
        D) if dataType=3
            i. the data extracted in this section describes the probability of this specific inelastic scattering reaction occurring using a simple interpolation scheme between energy and cross-section
            ii. [dqi]: should be the Qvalue for this reaction but instead they use the fsType for the reaction {eV}
            iii. [ilr]: break up flag, not used for anything
            iv. [total]: the number of incoming neutron kinetic energy, cross-section points
            v. for each energy cross-section point total, the following data is extracted
                a) [x]: the energy that the cross-section was measured at in {eV}
                b) [y]: the cross-section value in {barns}
        E) if dataType=4
            i. The data extracted in this section defines the probability that a hadron with a particular angle will be produced by an incoming neutron with a particular energy if this inelastic scattering reaction occurs
            ii. [theAngularDistributionType]: determines how the angular distribution of the produced hadrons are to be represented and subsequently the format of the data to be extracted
            iii. [targetMass]: the mass of the isotope {/(mass of a neutron)}
            iv. [frameFlag]: the reference frame that the data was measured with respect to (1=LAB, 2=CMS)
            v. if theAngularDistributionType = 0
                the angular distribution of the produced hadrons is to be represented by an isotropic distribution which means that no further data is necessary
            vi. if theAngularDistributionType = 1
                a) The angular distribution of the produced hadrons (remember the incoming is absorbed) is to be represented by Legendre polynomial sets which represent the probability angular relationship. There is a set of Legendre polynomials for each incoming neutron kinetic energy and each of the polynomials in a set are centered around a particular exit angle (meaning that the closer the angle is to the angle the Legendre polynomial is centered on, the better the approximation of that polynomial becomes). The coefficients of the Legendre polynomials are dependent on energy of the incoming neutrons and the angle of the exiting hadron.
                b) [nEnergy]: the number of incoming neutron kinetic energy points and also the number of Legendre

polynomial sets
    c) [nRanges]: defines the number of interpolation ranges for the interpolation of between the incoming neutron kinetic energy and the Legendre polynomial sets.
    d) For each range nRanges, the following data is extracted
        1. [range]: the end point of the range
        2. [it]: an integer representing the interpolation scheme to be used
    e) for each Legendre polynomial set nEnergy, the following data is extracted
        1. [temp]: the temperature of the isotope
        2. [energy]: the energy of the incoming neutron {eV}
        3. [tempdep]: not used for anything
        4. [nLegendre]: the number of Legendre polynomials in the set and also the number of exit angle probability points
        5. for each Legendre polynomial nLegendre, the following data is extracted
            i. [coeff]: the coefficient of the Legendre polynomial
            ii. note: the angle that the polynomial is centered around is based off the indices of the polynomial in the set
vii. if theAngularDistributionType = 2
    a) The angular distribution is represented by a 3D table of incoming neutron kinetic energy, exit angle and probability
    b) [nEnergy]: the number of incoming neutron kinetic energy points
    c) [nRanges]: defines the number of interpolation ranges for the interpolation of between the incoming neutron kinetic energy points.
    d) For each range nRanges, the following data is extracted
        1. [range]: the end point of the range
        2. [it]: an integer representing the interpolation scheme to be used
    e) for each incoming neutron kinetic energy point nEnergy, the following data is extracted
        1. [temp]: the temperature of the isotope that the data was taken at
        2. [energy]: the energy of the incoming neutron kinetic energy
        3. [tempdep]: not used for anything
        4. [neg]: the number of exit angle points for this incoming neutron kinetic energy
        5. [nRanges]: defines the number of interpolation ranges for the interpolation of between the exit angle points.
        6. For each range nRanges, the following data is extracted
            i. [range]: the end point of the range
            ii. [it]: an integer representing the interpolation scheme to be used
        7. for each exit angle point neg, the following data is extracted
            i. [eg]: the exit angle of the hadron
            ii. [pg]: the probability of this hadron being created

F) if dataType=5
    i. theEnergyDistribution->Init()
    ii. The data collected in this section defines the energy distribution of the produced hadrons using various models. Each model contains an incoming neutron kinetic energy and model weight (the model accuracy at a particular incoming neutron kinetic energy) interpolation scheme which allows the process to stochastically select a model to be used for a particular incoming neutron
    iii. [dummy]: does nothing
    iv. [theNumberOfPartials]: the number of partials that will make up interpolation scheme between incoming neutron kinetic energy and outgoing hadron energy
    v. for each partial theNumberOfPartials, the following data will be extracted
        a) [theRepresentationType]: Specifies how the data has been formatted
        b) if theRepresentationType=1
            1. [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
            2. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.
            3. For each range nRanges, the following data is extracted

i. [range]: the end point of the range
ii. [it]: an integer representing the interpolation scheme to be used

4. for each energy and representation weight point total, the following data is extracted
   i. [x]: the energy of the incoming neutron {eV}
   ii. [y]: the representation weight, aka the probability of this representation being used
5. [nDistFunc]: the number of interpolation schemes used to describe the relationship between out-going hadron energy and the probability of the hadron leaving with that energy. there is one interpolation scheme per incoming neutron kinetic energy so this is also the number of incoming neutron kinetic energy points
6. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and the energy probability of the ejected hadron.
7. For each range nRanges, the following data is extracted
   i. [range]: the end point of the range
   ii. [it]: an integer representing the interpolation scheme to be used
8. For each incoming neutron kinetic energy point nDistFunc, the following data is extracted
   i. [currentEnergy]: the incoming neutron kinetic energy {eV}
   ii. [total]: the number of outgoing hadron energy and probability of occurrence points
   iii. [nRanges]: defines the number of interpolation ranges for the interpolation between outgoing hadron energy and the probability of occurrence.
   iv. For each range nRanges, the following data is extracted
      a) [range]: the end point of the range
      b) [it]: an integer representing the interpolation scheme to be used
   v. For each out going hadron energy point total, the following data is extracted
      a) [x]: the out-going hadron energy {eV}
      b) [y]: the probability of occurrence

c) if theRepresentationType=5
1. For neatness the following is organised into three parts
2. theFractionalProb
   i. [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
   ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.
   iii. For each range nRanges, the following data is extracted
      a) [range]: the end point of the range
      b) [it]: an integer representing the interpolation scheme to be used
   iv. for each energy and representation weight point total, the following data is extracted
      a) [x]: the energy of the incoming neutron {eV}
      b) [y]: the representation weight, aka the probability of this representation being used
3. theThetaDist
   i. [total]: the total number of incoming neutron kinetic energy and average outgoing hadron energy points
   ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and average out-going neutron energy.
   iii. For each range nRanges, the following data is extracted
      a) [range]: the end point of the range
      b) [it]: an integer representing the interpolation scheme to be used
   iv. for each incoming neutron kinetic energy and maximum out-going hadron energy point total, the following data is extracted
      a) [x]: the incoming neutron kinetic energy {eV}
      b) [y]: the average out-going hadron energy
4. theXDist
   i. [total]: the number of multiplier and weight points. The weights are the probability of the multiplier being randomly selected. When an outgoing hadron energy is called for, the average outgoing hadron energy is interpolated from the ThetaDist data and a randomly selected multiplier is applied to it.

ii. [nRanges]: defines the number of interpolation ranges for the interpolation between multiplier and weight points.
iii. For each range nRanges, the following data is extracted
a) [range]: the end point of the range
b) [it]: an integer representing the interpolation scheme to be used
iv. for each energy and representation weight point total, the following data is extracted
a) [x]: the multiplier
b) [y]: the weight

d) if theRepresentationType=7,9
1. theFractionalProb
i. [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.
iii. For each range nRanges, the following data is extracted
a) [range]: the end point of the range
b) [it]: an integer representing the interpolation scheme to be used
iv. for each energy and representation weight point total, the following data is extracted
a) [x]: the energy of the incoming neutron {eV}
b) [y]: the representation weight, aka the probability of this representation being used
2. theThetaDist
i. [total]: the total number of incoming neutron kinetic energy and average outgoing hadron energy points
ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and average out-going neutron energy.
iii. For each range nRanges, the following data is extracted
a) [range]: the end point of the range
b) [it]: an integer representing the interpolation scheme to be used
iv. for each incoming neutron kinetic energy and maximum out-going hadron energy point total, the following data is extracted
a) [x]: the incoming neutron kinetic energy {eV}
b) [y]: the average out-going hadron energy {eV}

e) if theRepresentationType=11
1. theFractionalProb
i. [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.
iii. For each range nRanges, the following data is extracted
a) [range]: the end point of the range
b) [it]: an integer representing the interpolation scheme to be used
iv. for each energy and representation weight point total, the following data is extracted
a) [x]: the energy of the incoming neutron {eV}
b) [y]: the representation weight, aka the probability of this representation being used
2. theApar
i. [total]: the total number of incoming neutron kinetic energy and parameter A of the Watt Spectrum points
ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and parameter A of the Watt Spectrum.
iii. For each range nRanges, the following data is extracted
a) [range]: the end point of the range
b) [it]: an integer representing the interpolation scheme to be used
iv. for each incoming neutron kinetic energy and parameter A of the Watt Spectrum point total, the

following data is extracted
    a) [x]: the incoming neutron kinetic energy {eV}
    b) [y]: the parameter A of the Watt Spectrum
3. theBpar
    i. [total]: the total number of incoming neutron kinetic energy and parameter B of the Watt Spectrum points
    ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and parameter B of the Watt Spectrum.
    iii. For each range nRanges, the following data is extracted
        a) [range]: the end point of the range
        b) [it]: an integer representing the interpolation scheme to be used
    iv. for each incoming neutron kinetic energy and parameter A of the Watt Spectrum point total, the following data is extracted
        a) [x]: the incoming neutron kinetic energy {eV}
        b) [y]: the parameter B of the Watt Spectrum

f) if theRepresentationType=12
1. theFractionalProb
    i. [total]: the number of energy and representation weight points in this interpolation scheme. The representation weight refers to how well this representation defines what is experimentally observed at this incoming neutron kinetic energy. Thus the higher the probability the more likely this representation is to be used instead of the others.
    ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and representation weight points.
    iii. For each range nRanges, the following data is extracted
        a) [range]: the end point of the range
        b) [it]: an integer representing the interpolation scheme to be used
    iv. for each energy and representation weight point total, the following data is extracted
        a) [x]: the energy of the incoming neutron {eV}
        b) [y]: the representation weight, aka the probability of this representation being used
2. [theAvarageKineticPerNucleonForLightFragments]: the average kinetic energy of a light fragment per nucleon in the fragment {eV}
3. [theAvarageKineticPerNucleonForHeavyFragments]: the average kinetic energy of a heavy fragment per nucleon in the fragment {eV}
4. theMaxTemp
    i. [total]: the total number of incoming neutron kinetic energy and the max fission fragment temperature points
    ii. [nRanges]: defines the number of interpolation ranges for the interpolation between incoming neutron kinetic energy and max fission fragment temperature.
    iii. For each range nRanges, the following data is extracted
        a) [range]: the end point of the range
        b) [it]: an integer representing the interpolation scheme to be used
    iv. for each incoming neutron kinetic energy and parameter A of the Watt Spectrum point total, the following data is extracted
        a) [x]: the incoming neutron kinetic energy {eV}
        b) [y]: the max fission fragment temperature

G) if dataType=6
i. The data in this section describes the energy-angular distribution of the outgoing neutrons
ii. [targetMass]: the mass of the isotope {/(mass of a neutron)}
iii. [frameFlag]: the relativistic frame that the data is taken with respect to
iv. [nProducts]: the number of states that the nucleus can go into after the absorption of the neutron
v. for each states of the nucleus [nProducts], the following data is extracted
    a) [theMassCode]: the 5 digit ZZAAA code for the out going particle code
    b) [theMass]: not used for anything
    c) [theIsomerFlag]: not used for anything
    d) [theDistLaw]: specifies the distribution of the neutrons, i.e. isotropic, continuous angular-energy dist, discrete 2 body scattering
    e) [theGroundStateQValue]: not used for anything, the energy difference between is excited state and the

ground state, i.e. the theoretical Q-value of the reaction {eV}

f) [theActualStateQValue]: the energy difference between the excited state after absorption and the ground state before absorption, i.e. the actual Q-value of the reaction, should be very close to the theoretical {eV}
g) [total]: the total number of incoming neutron kinetic energy and out-going neutron multiplicity (the number of neutrons released) points.
h) [nRanges]: defines the number of interpolation ranges for the interpolation between the incoming neutron kinetic energy and out-going neutron multiplicity.
i) For each range [nRanges], the following data is extracted
    1. [range]: the end point of the range
    2. [it]: an integer representing the interpolation scheme to be used
j) for each out-going neutron multiplicity point [total], the following data is extracted
    1. [x]: the incoming neutron kinetic energy {eV}
    2. [y]: the out-going neutron multiplicity
k) if theDistLaw=0,3
    1. An isotropic neutron angular distribution is used
    2. No data is collected from the file, instead uniform random distributions are used to sample the angles
l) if theDistLaw=1
    1. A continuous energy and angular distribution is used
    2. [theTargetCode]: is the ZZAAA integer that specifies the nucleus
    3. [theAngularRep]: is an integer that specifies how the angular distribution is to be sampled
    4. [theInterpolation]: specifies the interpolation scheme to be used for the interpolation between probability of the neutron being created with a certain energy and angle, and the incoming neutron kinetic energy
    5. [nEnergy]: the number of incoming neutron kinetic energy points
    6. for each incoming neutron kinetic energy point [nEnergy], the following data is extracted
        i. [theEnergy]: the incoming neutron kinetic energy {eV}
        ii. [nEnergies]: the number of neutron energy points
        iii. [nDiscreteEnergies]: the number of neutron energy points that have discrete energies i.e. the number of points that are not part of the continuous neutron energy dist (the neutron can only occur at the energy point, there is no interpolation in-between the points), thus nDiscreteEnergies<= nEnergies, and nEnergies- nDiscreteEnergies is the number of neutron energy points that are part of the continuous energy dist.
        iv. [nAngularParameters]: the number angle probability points for each neutron energy point
        v. for each neutron energy point [nEnergies], the following data is extracted
            a) [sEnergy]: the out-going neutron energy {eV}
            b) for each angle probability point nAngularParameters, the following data is extracted
                1. [y]: the probability of the occurrence of the neutron (the neutron meaning at the specified isotope state, incoming neutron kinetic energy, neutron energy, and angle). Note the angle corresponding to the probability is based off the probability array index, i.e. each array index in the probability array is a set angle apart.
m) if theDistLaw=2,4
    1. a discrete 2 body recoil distribution is used, this type of distribution uses a set of Legendre polynomials to describe the angular dependence of the probability of a neutron occurring for each neutron energy, there are multiple neutron energies for each incoming neutron kinetic energy
    2. [nEnergy]: the number of neutron energy points
    3. [nRanges]: defines the number of interpolation ranges for the interpolation of between the Legendre polynomial sets.
    4. For each range [nRanges], the following data is extracted
        i. [range]: the end point of the range
        ii. [it]: an integer representing the interpolation scheme to be used
    5. for each neutron energy point [nEnergy], the following data is collected
        i. [energy]: the incoming neutron kinetic energy {eV}
        ii. [aRep]: the interpolation scheme between the Legendre coefficients
        iii. [nCoeff]: the number of Legendre polynomial coefficients
        iv. if aRep>0
            a) then the total number of Legendre coefficients is 2xnCoeff

b) for the total number of Legendre coefficients, the following data is extracted
1. [y]: is Legendre polynomial coefficient

v. if aRep=0 then
a) then the total number of Legendre coefficients is nCoeff
b) for the total number of Legendre coefficients, the following data is extracted
1. [y]: is Legendre polynomial coefficient

n) if theDistLaw=5
1. does nothing, has not been coded yet

o) if theDistLaw=6
1. an n-body phase space distribution is used
2. [theTotalMass]: the mass of the isotope divided by the mass of a neutron {MeV}
3. [theTotalCount]: the number of particles involved in the reaction

p) if theDistLaw=7
1. an experimental parameterized energy and angular model
2. [nEnergies]: the number of incoming neutron energies
3. [nRanges]: defines the number of interpolation ranges for the interpolation of between incoming neutron kinetic energy and the probability of the neutron being shot in at a specific angle.
4. For each range [nRanges], the following data is extracted
i. [range]: the end point of the range
ii. [it]: an integer representing the interpolation scheme to be used
5. for each incoming neutron kinetic energy [nEnergies], the following data is collected
i. [theEnergies]: the incoming neutron kinetic energy {eV}
ii. [nCosTh]: the number of angle points for this incoming neutron kinetic energy
iii. [nRanges]: defines the number of interpolation ranges for the interpolation of between the neutron angular direction and the probability of the neutron having a specific energy.
iv. For each range [nRanges], the following data is extracted
a) [range]: the end point of the range
b) [it]: an integer representing the interpolation scheme to be used
v. for each angle point [nCosTh], the following data is extracted
a) [label]: this is the cosine of the angle
b) [total]: the number of neutron energy points
c) [nRanges]: defines the number of interpolation ranges for the interpolation of between the neutron energy and the probability of the neutron occurring.
d) For each range nRanges, the following data is extracted
1. [range]: the end point of the range
2. [it]: an integer representing the interpolation scheme to be used
e) for each neutron energy point [total], the following information is extracted
1. [x]: the neutron energy {eV}
2. [y]: the probability of occurrence

H) if dataType=12
i. defines the relationship between outgoing photon energy, the probability of the photon occurring and the fraction of photons that will transfer their kinetic energy to an electron
ii. The first number [repFlag] refers to the format that the data is presented in
iii. if repFlag can be extracted
a) The second number [targetMass] is the mass of the isotope in {/(mass of a neutron)}
b) Now depending on the repFlag the rest of the data is extracted in the following ways
c) repFlag=1
1. this format uses a simple interpolation scheme to represent the dependence of photon multiplicity on the incoming neutron kinetic energy, for each possible discrete photon energy
2. the next number [nDiscrete] is the number of discrete types of photons that can be emitted for this isotope
3. For each discrete type of photon the following data is collected
i. [disType]: the photon energy distribution type (continuous [1] or discrete [0]) , determines whether the photon energy will be interpolated or not
ii. [energy]: the energy of the produced photon in {eV}
iii. [total]: the number of incoming neutron kinetic energy points
iv. [nRanges]: defines the number of interpolation ranges (see [1] for explanation of ranges) for the

interpolation of incoming neutron kinetic energy and the photon multiplicity.
v. For each range [nRanges], the following data is extracted
a) [range]: the end point of the range
b) [it]: an integer representing the interpolation scheme to be used
vi. For each incoming neutron kinetic energy point [total], the following data is extracted
a) [x]: the incoming neutron kinetic energy {eV}
b) [y]: the photon multiplicity (i.e. the number of photons produced)

d) repFlag=2
1. This format is a table of photon energy vs the relative probability of the photon being produced at that energy. The effects of different incoming neutron energies on the photon energy are taken care of later in the application of the process to the track
2. [theInternalConversionFlag]: this number isn't used for anything
3. [theBaseEnergy]: is the base photon energy used in the creation of photons by the inelastic process {eV}
4. [theInternalConversionFlag]: specifies whether the photon transition fraction data is present in the file
5. [nGammaEnergies]: specifies the number of photon energy points that there is data for
6. For each photon energy point [nGammaEnergies], the following data is collected
i. if theInternalConversionFlag = 1
a) [theLevelEnergies]: The energy difference between the isotope before the reaction (ground state) and after the photon has been produced (some excited state) {eV}. Photon energy = theBaseEnergy- theLevelEnergies
b) [theTransitionProbabilities]: The probability of a photon with this energy being produced
ii. if theInternalConversionFlag = 2
a) [theLevelEnergies]: The energy difference between the isotope before the reaction (ground state) and after the photon has been produced (some excited state) {eV}. Photon energy = theBaseEnergy- theLevelEnergies
b) [theTransitionProbabilities]: The probability of a photon with this energy being produced
c) [thePhotonTransitionFraction]: The probability of an electron with the same KE as the photon being released instead of the photon (due to the absorption of the photon by a valence electron)

I) if thedataType=13
i. the data collected in this section uses a 3D table to describe the probability of a photon with a particular energy being created for various incoming neutron energies, this information is only used in the case that repFlag=0, currently this section of the code is not meant to be used, setting repFlag to 0 will cause an error
ii. [nDiscrete]: the number of photon energy points
iii. [targetMass]: the mass of the isotope
iv. theTotalXsec
a) None of the data in this section is used for anything
b) [total]: the number of incoming neutron kinetic energy and total cross-section points
c) [nRanges]: the number of ranges
d) For each range nRanges, the following data is extracted
1. [range]: the end point of the range
2. [it]: an integer representing the interpolation scheme to be used
e) for each incoming neutron kinetic energy and total cross-section point total, the following is extracted
1. [x]: the incoming neutron kinetic energy {eV}
2. [y]: the total cross-section {barn}
v. for each photon energy point nDiscrete, the following data is extracted
a) [theGammas[i]]: the energy of the photon {eV}
b) [theShells[i]]: not used for anything {eV}
c) [isPrimary[i]]: not used for anything
d) [disType[i]]: specifies whether the distribution is discrete or continuous, (if it is discrete there is no interpolation between points)
e) thePartialXsec[i]
1. [total]: the number of incoming neutron kinetic energy and cross-section points
2. [nRanges]: the number of ranges used to interpolate between the incoming neutron kinetic energy and the cross-section of the reaction producing a photon with energy theGammas[i]

3. For each range nRanges, the following data is extracted
    i. [range]: the end point of the range
    ii. [it]: an integer representing the interpolation scheme to be used
4. for each incoming neutron kinetic energy and cross-section point total, the following is extracted
    i. [x]: the incoming neutron kinetic energy {eV}
    ii. [y]: the cross-section {barn}

J) if the dataType =14
i. Defines the relationship between incoming neutron kinetic energy and the probability of a neutron with a specific angle and energy being produced
ii. The next number [isoFlag] specifies whether the distribution of photons is isotropic or not
iii. if isoFlag=1
a) No more data is collected, the momentum direction angles of created photons will be selected from a uniformly random distribution.
iv. If isoFlag!=1
a) if repFlag==2
1. The code spits out an error since this should not occur
2. [tabulationType]: specifies how the photon angular probability is tabulated, either as Legendre polynomials, or a simple interpolation scheme (in both cases the probability of occurrence is a function of the cosine of the angle)
3. [nDiscrete2]: is the number of discrete photon energies that can be emitted for this isotope, should be equal to nDiscrete
4. [nIso]: is the number of isotropic photons (i.e.: photons where the probability is independent of angle)
5. for each isotropic photon i.e. [nIso], the following data is collected
i. [theGammas]: The energy of the released photon {eV}
ii. [theShells]: The binding energy of the neutron for a particular shell, this array is not used for anything {eV}
6. for each nonisotropic photon [nDiscrete-nIso], the following data is collected
i. If tabulationType==1 (i.e. photon angular probability is tabulated as Legendre polynomials which depend on the cosine of the direction angle)
a) [theGammas]: The energy of the released photon {eV}
b) [theShells]: The binding energy of the neutron for a particular shell, this array is not used for anything {eV}
c) [nNeu]: the number of incoming neutron kinetic energy points for this photon
d) The data in this section is extracted but not used for anything
1. [nRanges]: defines the number of interpolation ranges (see [1] for explanation of ranges) for the interpolation of incoming neutron kinetic energy and the Legendre polynomials that relate probability of occurrence to the scattering angle
2. For each range [nRanges], the following data is extracted
[range]: the end point of the range
[it]: an integer representing the interpolation scheme to be used
e) for each incoming neutron kinetic energy point [nNeu], the following data is collected
1. [eNeu]: the energy of the incoming neutron {eV}
2. [nPoly]: the number of the Legendre polynomials used to describe the relationship between cos(theta) (theta is the scattering angle and the probability of occurrence, for this particular case (the domain of a Legendre polynomial is cos(theta) and the range is the probability)
3. for each Legendre polynomial nPoly, the following data is collected
[coeff]: the coefficient of the Legendre polynomial
ii. If tabulationType=2 (i.e. photon angular probability is tabulated as a simple interpolation scheme)
a) [theGammas]: The energy of the released photon {eV}
b) [theShells]: The binding energy of the neutron for a particular shell, this array is not used for anything {eV}
c) [nNeu]: the number of incoming neutron kinetic energy points for this photon

d) for each incoming neutron kinetic energy point [nNeu], the following data is collected

1. [eNeu]: the energy of the incoming neutron {eV}
2. [nProb]: the number of probability points at this energy
3. [nRanges]: defines the number of interpolation ranges (see [1] for explanation of ranges) for the interpolation of cosTheta and probability of occurrence
4. For each range [nRanges], the following data is extracted
[range]: the end point of the range
[it]: an integer representing the interpolation scheme to be used
5. for each probability point [nProb], the following data is collected
[cosTheta]: the cosine of the direction angle of the photon
[probDist]: the probability of the photon being released under the previous circumstances

K) if the dataType =15
   i. In this section the data that defines the continous photon energy distributions is collected
   ii. if disType[i]!=1 for any of the photon energy points extracted in InitMean
      a) no more data is collected, the energy of created photons will not be interpolated and will simply be the closest discrete photon energy point
   iii. else
      a) [nPartials]: the number of distinct states that the nuclei will assume after absorbing the neutron
      b) for each distinct states that the nuclei will assume [nPartial], the following data is extracted
         1. [dummy]: useless number
         2. [total]: number of points for interpolating the probability of this state occurring
         3. [nRanges]: defines the number of interpolation ranges (see [1] for explanation of ranges) for the interpolation of incoming neutron kinetic energy and the probability of the nucleus entering this distinct state.
         4. For each range [nRanges], the following data is extracted
            i. [range]: the end point of the range
            ii. [it]: an integer representing the interpolation scheme to be used
         5. for each interpolation point [total], the following data is extracted
            i. [x]: the incoming neutron kinetic energy {eV}
            ii. [y]: the probability of occurrence of this state at the incoming neutron kinetic energy
         6. [nen]: number of incoming neutron kinetic energy points for the interpolation of the photon probability distribution
         7. [nRanges]: defines the number of interpolation ranges for the interpolation of incoming neutron kinetic energy and the probability of the photon being created with a specific energy.
         8. For each range [nRanges], the following data is extracted
            i. [range]: the end point of the range
            ii. [it]: an integer representing the interpolation scheme to be used
         9. For each neutron energy point [nen], the following data is extracted
            i. [e]: the incoming neutron kinetic energy {eV}
            ii. [neg]: the number of photon energy points
            iii. [nRanges]: defines the number of interpolation ranges for the interpolation of the photon energy and the probability of the photon occurring.
            iv. For each range [nRanges], the following data is extracted
               a) [range]: the end point of the range
               b) [it]: an integer representing the interpolation scheme to be used
            v. For each photon energy point [neg], the following data is extracted
               a) [eg]: the photon energy {eV}
               b) [pg]: the probability of occurrence

4. Now the data from the appropriate file in the /Inelastic/Gammas/ directory is extracted to describe the produced photons from the reaction, this data is only used if secondary photon data defined above is not supplied (dataType 12-15)
   A) The data collected in this section defines the energy distribution of the photons released by the radioactive decay of the isotope after inelastic scattering has occurred
   B) While there is data left in the file

i. [levelEnergy]: the excitation energy of the isotope {keV}
ii. if levelEnergy was successfully extracted
    a) [gammaEnergy]: the energy of the produced photon {keV}
    b) [probability]: the probability of a photon with energy gammaEnergy being produced when the isotope has levelEnergy excitation energy

[1]: A range is a set of data with an assigned interpolation scheme. An example of an interpolation vector with three ranges would be a vector with the first third of its data using a linear fit the second third of its data using a polynomial fit and the last third of its data as a linear fit.

[2]: the angular distribution of the produced hadrons (remember the incoming is absorbed) is to be represented by Legendre polynomials which represent the probability angular relationship; the coefficients of the Legendre polynomials are dependent on energy of the incoming neutrons and the angle of the exiting hadron

[3]: A partial seems to be very similar to a range, instead of specifying an interpolation scheme a distribution is assigned to each partial. Each partial covers a part of the domain, the combination of all the partials covers the whole domain.

[4]: incident neutron energy is always with respect to lab frame

Possible issues with NeutronHP

1. G4NeutronHPInelasticBaseFS.cc: lines 257-266, the theEnergyAngData should be checked first to ensure the target and neutron is set when it is checked first and used in the block below